

Geometric Realization of Bicomplex Signal Manifolds: Spectral Stability, Whitney Folds, and Monodromy of the Observation Field

Akari Hayami (Jian-Yu Huang)

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Abstract

We study a geometric framework for the realization of signal manifolds in bicomplex space $\mathbb{C}^2$. Starting from the momentum map $\mu : \mathbb{C}^2 \rightarrow \mathbb{R}^2$ and the power-conservation constraint $|z_1|^2 + |z_2|^2 = 1$, we derive a ruled-surface immersion $S(t, u) \in \mathbb{R}^3$ whose singularity structure is entirely governed by the quadratic invariant $\mathcal{P}(c) = c^2 - 3c + 1$, with unique root $c_0 = \varphi^{-2}$ in $(0, 1)$.

The paper separates intrinsic geometry from signal-theoretic and physical observations. The following results are proved rigorously. *(i)* The two lateral interior rank-loss points $P_{\pm}$ are the non-degenerate interior zeros of the complex observation field $F = N + iM$ and carry topological indices $\text{Ind}(P_{\pm}) = \mp 1$, with $|\det(DF)|_{P_{\pm}}| = 5s_0(1 - c_0)^2/Q_0$. The remaining boundary rank-loss point P_0 is a boundary half-cross-cap for the immersion and a degree-zero fold for the planar observation field (Theorem 4.1, Corollary 4.4, Proposition 8.8). *(ii)* Any locally defined square root $\chi = \sqrt{F}$ acquires a sign reversal after one loop around P_+ , requiring 4π for geometric restoration — a monodromy result derived solely from the topological index (Theorem 4.2). *(iii)* The derivative of $\mathcal{P}$ at its root gives $|\mathcal{P}'(c_0)| = \sqrt{5}$ exactly (Lemma 5.1). *(iv)* The Mellin dilation operator $\mathcal{T}_{\sigma} = -i(x\partial_x + \sigma)$ is formally self-adjoint iff $\sigma = \frac{1}{2}$, with adjoint $\mathcal{T}_{\sigma}^* = \mathcal{T}_{1-\sigma}$ derived by integration by parts (Theorem 7.1). When $T > 2(M - 1)/\delta$, the associated phasor family is linearly independent with condition number $\kappa(G) \leq (1 + \varepsilon)/(1 - \varepsilon)$ (Proposition 7.6, Corollary 7.8). *(v)* The fold and cross-cap normal forms have the intrinsic sublevel exponent $3/4$; this yields a parameter-free logarithmic resolution law but not a Shannon capacity formula. *(vi)* The normalization map of a cross-cap produces a canonical constructible defect sheaf supported on its open double locus. *(vii)* The Gram limit is the Haar orthogonality relation of the Bohr torus, while $\sigma = \frac{1}{2}$ is independently singled out by the Dirichlet–Hardy evaluation boundary. We also prove three scope limitations: the proposed mass ratio retains a length scale, the natural sign-holonomy boundary Dirac operator has vanishing APS η -invariant, and the constrained source surface is not invariant under the full ambient $SU(2)$ action. *(viii)* After explicitly complexifying the cross-cap residual, its germ is analytically equivalent to $X^2 + Y^4$; its Milnor fiber is a twice-punctured genus-one curve with monodromy spectrum $\{-i, 1, i\}$ and Deligne weight-graded dimensions $(2, 1)$ in weights one and two. *(ix)* Every positive-radius link of the standard real cross-cap is a circle whose image has one transverse double point and is therefore a figure-eight graph. Its normalization lift in $\mathbb{R}^4$ is embedded. The primitive weighted phase action of $X^2 + Y^4$ has weights $(2, 1)$ and produces an embedded orbit in $\mathbb{C}^2$ whose rectangular shadow is a transverse Lissajous figure-eight with exact dimensionless frequency ratio $2 : 1$. *(x)* For the actual lateral ruled-surface germs $P_{\pm}$, the three canonical Whitney two-jet invariants $(a_{\pm}, b_{\pm}, c_{\pm})$ satisfy $b_{\pm}^2 + a_{\pm}c_{\pm}^2 = 0$. Hence the normal-plane trace of the associated principal-curvature surface is a cusp rather than a transverse figure-eight;

this closes that direct geometric bridge. *(xi)* On the regular source, the induced Euclidean metric supplies a canonical scalar L^2 metric and Dirichlet form. Every nested conforming Galerkin refinement then obeys exact mass- and stiffness-matrix pullback identities. This fixes the normalized kinetic operator relative to the embedded metric. Each rank-loss point has zero local Dirichlet capacity, and, after fixing the outer Dirichlet boundary, deleting P_+ and P_- does not change the closed Friedrichs form. Uniform estimates on the front and seam charts give one logarithmic Green channel at each interior cross-cap. The minimal punctured Laplacian has deficiency indices $(2, 2)$, its Green pairing is explicit, and its self-adjoint extensions form a $U(2)$ family. Geometry alone does not select a non-Friedrichs member of that family. If one independently requires a submarkovian diffusion or a reference-scale-invariant boundary plane with no running coupling, the Friedrichs member is unique. Globally, the fixed-boundary operator is strictly positive, and its zero-energy Green matrix selects the Krein–von Neumann extension as a canonical non-Markovian member. Its nonzero spectrum has an intrinsic buckling variational principle. The direct normalization-to-Green bridge is also resolved negatively: extension by zero has zero pinch stalk, and its nonzero nearby/costalk line is sheet-odd, whereas the unique scalar logarithmic Green channel is sheet-even. Thus no local sheet-equivariant morphism exists without adding a sign-twisted target or an odd vector-valued defect channel. An arbitrary global multiplier of a classical action remains unfixed. The already proved square-root monodromy does canonically supply a flat sign line on each punctured source neighbourhood. In the exact cone normal family its scalar connection Laplacian has two lowest half-integer channels, whereas the disk-bounding spin Dirac operator twisted by the same line has one zero angular channel per chiral block. A direct front/seam parametrix promotes this to a one-dimensional actual chiral domain quotient; the full rank-two Dirac quotient is two-dimensional. Identification with the distinct normalization-deck sign line is obstructed already at first order: the two branch phases lift to a nonsplit $\mathbb{Z}_4$ torsor, and normalization-sheet reflection exchanges the two chiral trace lines. The observation field does fix the kernel-line norm. On the normalization cut, the Dirac Green form then derives a full-spin transmission action: its real transgression coefficient is $1/2$, the relative sheet coupling has unit modulus, and Pin compatibility reduces it to the two central signs. The overall action multiplier remains unfixed.

The repeated appearances of $\sqrt{5}$, the 4π restoration, and the half-density line remain noteworthy signal or physical analogies. They are reported as observations rather than as established capacity, particle-mass, chirality, or Weil-cohomological identifications. The Deligne structure just stated belongs only to the explicitly chosen complex residual fiber; it is not inferred from $\sigma = 1/2$ or from the real signal geometry alone.

1 Introduction

This study investigates a geometric model connecting signal physics and singularity theory using the tools of classical differential geometry. We define a signal manifold $\mathcal{S} \subset \mathbb{C}^2$ and an associated operator space $\mathcal{H}_S \cong L^2([1, \infty), dx)$ equipped with a weighted Hilbert metric:

$$\langle f, g \rangle_\sigma = \int_1^\infty f(x) \overline{g(x)} x^{2\sigma-1} dx.$$

Two singular structures must be distinguished. Rank loss of $S : D \rightarrow \mathbb{R}^3$ produces cross-cap germs, whereas the planar observation map $\Psi = (N, M) : D \rightarrow \mathbb{R}^2$ has fold behaviour. We show that the topological dipole of $F = N + iM$ implies a 4π monodromy (Section 4), and that the formal-adjoint threshold of the associated dilation operator occurs at $\sigma = \frac{1}{2}$ (Section 7). Section 8 replaces earlier condensed/Weil analogies by constructible-sheaf, Hardy–Dirichlet, and Bohr–Haar statements with explicit scope limits. It also identifies the pullback Euclidean metric as the canonical source-side Hilbert and Dirichlet datum, while separating this normalized

operator statement from an unfixed global action multiplier. Its isolated rank-loss points are removable for the fixed-boundary Friedrichs form. A front/seam parametrix now identifies one logarithmic defect channel at each interior cross-cap and classifies the resulting Green boundary space; Markov uniqueness and a separate no-running-scale argument both select only the Friedrichs member locally. A separate global positivity argument and the zero-energy Weyl matrix select the canonical Krein–von Neumann extension, without identifying it with the singular-image quotient. A sheaf-theoretic audit now proves why that identification fails for the present scalar operator: the normalization defect has zero pinch stalk, and its nearby line transforms by the sheet-sign representation while the logarithmic Green channel is even. The same audit now separates the real cross-cap residual from its explicitly chosen complex algebraic completion, for which Milnor monodromy and Deligne weights can be computed. The same completion also supplies a primitive $(2, 1)$ phase lift: its smooth four-real-dimensional orbit projects to a planar figure-eight, while one restored phase coordinate gives an exact reconstruction. A separate two-jet calculation for the actual lateral ruled-surface cross-caps shows that their principal-curvature exceptional traces remain cusps and therefore do not identify either figure-eight construction. The regular source is orientable and has boundary; the local sheet exchange is a normalization defect, not the orientation class of a closed non-orientable surface. Signal and physical similarities are retained as noteworthy observations, never as substitutes for missing operators or dynamics.

2 Preliminaries

2.1 Bicomplex Space and Unitary Modulations

Let $\mathbb{C}^2 \cong \mathbb{R}^4$. A signal state $\mathbf{z} = (z_1, z_2)^\top$ under the total power constraint $|z_1|^2 + |z_2|^2 = 1$ defines a 3-sphere S^3 . Unitary signal modulations correspond to $SU(2)$ operations on this ambient space. The constrained two-dimensional source used below is a proper subset of S^3 and is not itself invariant under the full transitive left action of $SU(2)$; Proposition 8.25 makes this distinction precise.

2.2 Whitney Fold Singularities

A smooth mapping $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has a *fold singularity* at a point if it is locally equivalent to the normal form $(\xi, \eta) \mapsto (\xi, \eta^2)$ [4, 5, 3]. In the context of 4D-to-3D projections, this singularity generates a double-sheeted covering of the observational plane, where the two sheets represent distinct internal phases of the same observable coordinate.

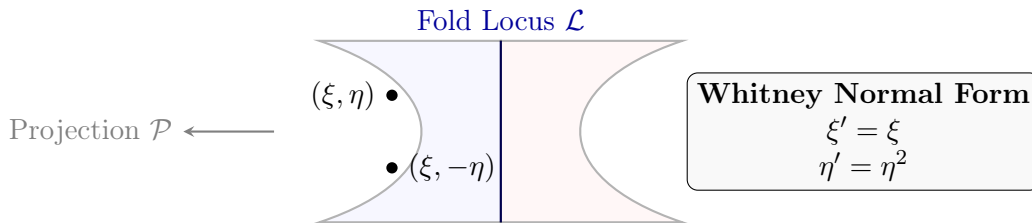


Figure 1: Schematic of a Whitney fold singularity. The mapping collapses two sheets onto a single observational plane, creating a $2 : 1$ covering.

3 Formal Definitions

3.1 The Bicomplex Signal Manifold

The Bicomplex Signal Manifold $\mathcal{S}$ is derived from the constraint $|z_1|^2 + |z_2|^2 = 1$ via the momentum map $\mu : \mathbb{C}^2 \rightarrow \mathbb{R}^2$, $\mu(\mathbf{z}) = (|z_1|^2, |z_2|^2) = (X, Y)$. Setting $X = c := \cos t$ and $Y = 1 - c = \cos \phi$, the constraint $\cos t + \cos \phi = 1$ parametrizes a family of ruling lines. On the source rectangle $D = [-\pi/2, \pi/2] \times [0, 1]$, with $Q(t) = \sqrt{2c - c^2}$ and ruling parameter $u \in [0, 1]$, the immersion is

$$S(t, u) = (c + 2u(1 - c), (1 - u)s, uQ)^\top, \quad s = \sin t, \quad (1)$$

a non-developable ruled surface studied in detail in [1]; basic differential-geometric notation follows [27].

Definition 3.1 (Geometric Projection). *The Geometric Projection $\mathcal{P} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ maps the 4D signal state to 3D observational space. The realization $\mathcal{P}(\mathcal{S})$ exhibits immersion singularities where the normal vector $V = S_t \times S_u$ vanishes.*

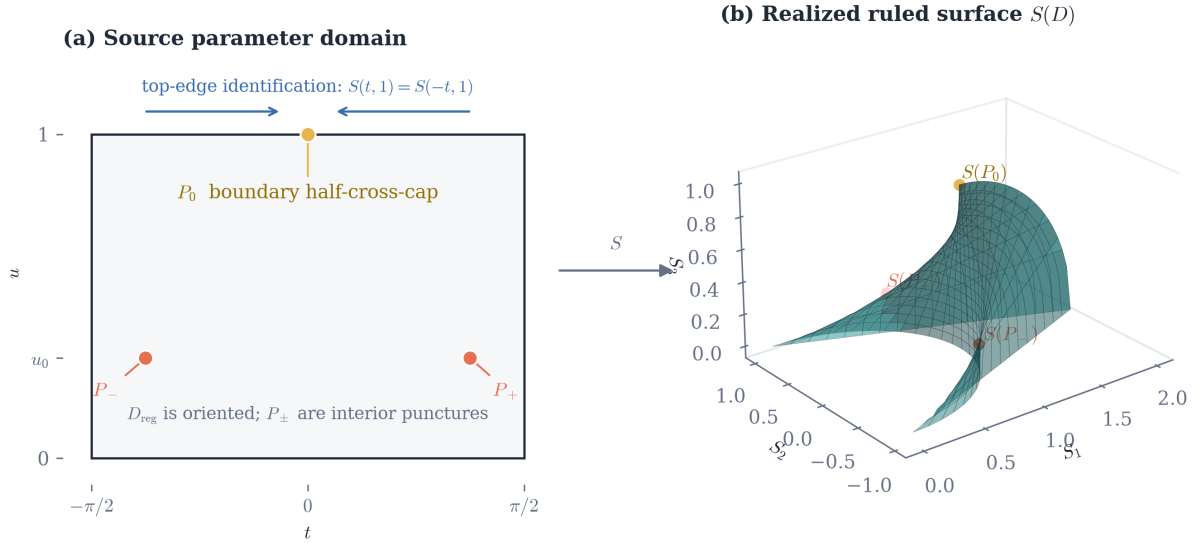


Figure 2: Source and realization atlas for the immersion (1). The two lateral interior points $P_\pm$ are the rank-loss cross-caps, while P_0 is the boundary half-cross-cap. The top-edge relation $S(t, 1) = S(-t, 1)$ has the fixed point P_0 ; it is not a free gluing that makes the source non-orientable. The right panel shows the corresponding realized ruled surface.

Definition 3.2 (Mellin–Hilbert Spectral Operator). *The Mellin–Hilbert Spectral Operator is the family of operators*

$$\mathcal{T}_\sigma = -i \left(x \frac{d}{dx} + \sigma \right), \quad \sigma \in \mathbb{R},$$

acting on $\mathcal{D} = C_c^\infty((1, \infty)) \subset L^2([1, \infty), dx)$. The formal eigenfunctions $f_\omega(x) = x^{-(\sigma+i\omega)}$ satisfy $\mathcal{T}_\sigma f_\omega = -\omega f_\omega$ with real eigenvalue $-\omega$ (see Corollary 7.3).

Definition 3.3 (Singular Resolution Cost). *For a non-negative local residual K and $\varepsilon > 0$, define*

$$V_K(\varepsilon) := \text{vol}\{x : K(x) \leq \varepsilon\}, \quad \mathcal{C}_K(\varepsilon) := -\log V_K(\varepsilon).$$

This is a geometric resolution cost. It is not called a communication-channel capacity unless a probability law, noise model, and resource constraint are specified independently.

4 Geometric Monodromy of the Observation Field

The cross product $V = S_t \times S_u$ yields the complex observation field $F := N + iM$ where $N = V_z$ and $M = V_x$. The following summarises results from [1].

Theorem 4.1 (Interior Topological Dipole). *The field $F = N + iM$ has exactly two non-degenerate isolated zeros in the interior of D , namely $P_{\pm} = (\pm t_0, u_0)$, with indices $\text{Ind}(P_+) = -1$ and $\text{Ind}(P_-) = +1$. At P_+ the Jacobian determinant of $(t, u) \mapsto (N, M)$ equals $-5s_0(1 - c_0)^2/Q_0 < 0$, confirming non-degeneracy. In addition, $F(P_0) = 0$ at the boundary point $P_0 = (0, 1)$; this zero is degenerate and has local degree zero by Proposition 8.8, so it is not a third member of the interior vortex-antivortex pair.*

Theorem 4.2 (Geometric Monodromy and 4π Restoration). *Let $\gamma : [0, 1] \rightarrow U \setminus \{P_+\}$ be a small positively oriented loop encircling P_+ once. Any locally defined square root $\chi = \sqrt{F}$ along γ satisfies*

$$\chi(\gamma(1)) = -\chi(\gamma(0)).$$

Consequently the geometric period of χ in parameter space is 4π : two full traversals of γ are required to restore χ .

Proof. By Theorem 4.1, $\text{Ind}(P_+) = -1$, meaning the winding number of $F \circ \gamma$ around the origin in $\mathbb{C}$ equals -1 . Write $F(\gamma(s)) = R(s)e^{i\alpha(s)}$ for a continuous argument α . Since the winding number is -1 , we have $\alpha(1) = \alpha(0) - 2\pi$.

Define the square root branch $\chi(\gamma(s)) = \sqrt{R(s)}e^{i\alpha(s)/2}$. At $s = 1$:

$$\begin{aligned} \chi(\gamma(1)) &= \sqrt{R(0)}e^{i(\alpha(0)-2\pi)/2} \\ &= \sqrt{R(0)}e^{i\alpha(0)/2} \cdot e^{-i\pi} \\ &= -\chi(\gamma(0)). \end{aligned}$$

After a second traversal, $\alpha(2) = \alpha(0) - 4\pi$, giving $\chi(\gamma(2)) = \sqrt{R(0)}e^{i(\alpha(0)-4\pi)/2} = +\chi(\gamma(0))$. Thus 4π is the minimal period.

The sign flip $\chi \rightarrow -\chi$ is the deck transformation of the phase square-root cover. It must not yet be identified with exchange of the two normalization sheets of a Whitney cross-cap. The latter approach the pinch in opposite kernel directions, so their limiting F -phases are antipodal and their square roots differ by a quarter-turn, not by a sign. This distinction is made precise in Theorem 8.22. $\square$

Remark 4.3. *Theorem 4.2 establishes a purely geometric monodromy result: the complex field F , whose topological dipole structure is rigorously derived from the surface geometry, forces any branch of $\sqrt{F}$ to have period 4π . This is analogous to spinor periodicity in $SU(2)$, but the identification with physical $\text{Spin}^{-\frac{1}{2}}$ requires additional physical assumptions and is not claimed here as a mathematical theorem.*

Corollary 4.4 (Local Degree via Jacobian Sign). *For a smooth map $\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with an isolated zero at p and non-degenerate Jacobian $\det(D\Phi(p)) \neq 0$, the topological index satisfies*

$$\text{Ind}(p) = \text{sgn}(\det(D\Phi(p))).$$

Applied to $\Phi = (N, M)$ and the lateral singularities:

$$\begin{aligned} \det(DF)|_{P_+} &= -\frac{5s_0(1-c_0)^2}{Q_0} < 0 \Rightarrow \text{Ind}(P_+) = -1, \\ \det(DF)|_{P_-} &= +\frac{5s_0(1-c_0)^2}{Q_0} > 0 \Rightarrow \text{Ind}(P_-) = +1. \end{aligned}$$

This gives an explicit, computable derivation of the topological dipole, with numerical values $|\det(DF)|_{P_{\pm}}| \approx 2.245$.

Proof. The local degree formula $\text{Ind}(p) = \text{sgn}(\det D\Phi(p))$ is standard for non-degenerate zeros of smooth planar maps [4]. The Jacobian values are computed in [1] (Theorem 4.1) from the explicit formulae N_u, N_t, M_u, M_t at (c_0, u_0) using $c_0^2 = 3c_0 - 1$. The parity identity $\det(DF)|_{P_-} = -\det(DF)|_{P_+}$ follows from N, M being even/odd functions of t respectively, so the sign flips under $t_0 \mapsto -t_0$. $\square$

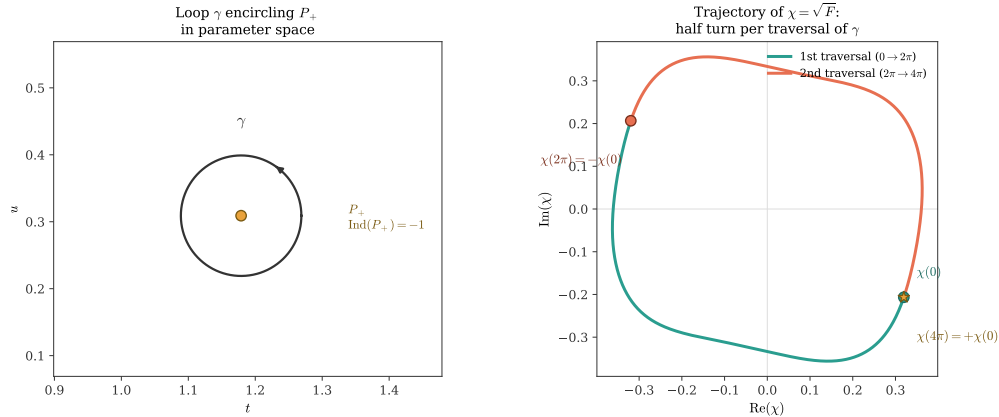


Figure 3: Geometric monodromy of $\chi = \sqrt{F}$ (Theorem 4.2). Left: a positively oriented loop γ encircling P_+ in parameter space; $\text{Ind}(P_+) = -1$. Right: the resulting trajectory of χ in $\mathbb{C}$. Because the index is -1 , the argument of χ advances at half the rate of γ 's winding, so one traversal (teal) sweeps only half a turn, from $\chi(0)$ to the antipodal point $\chi(2\pi) = -\chi(0)$; a second traversal (coral) is required to close the loop, giving the minimal geometric period 4π .

5 Intrinsic Singular Compression at the Anchor

We first retain the exact derivative attached to the normalized invariant polynomial, then derive a coordinate-invariant local scaling exponent from the singularity normal form.

Lemma 5.1 (Spectral Compression Rate). *The derivative of the core invariant polynomial $\mathcal{P}(c) = c^2 - 3c + 1$ at its root $c_0 = (3 - \sqrt{5})/2$ satisfies*

$$|\mathcal{P}'(c_0)| = \sqrt{5}.$$

Proof. $\mathcal{P}'(c) = 2c - 3$. At $c_0 = (3 - \sqrt{5})/2$:

$$\mathcal{P}'(c_0) = 2 \cdot \frac{3 - \sqrt{5}}{2} - 3 = (3 - \sqrt{5}) - 3 = -\sqrt{5}.$$

Hence $|\mathcal{P}'(c_0)| = \sqrt{5}$. □

Remark 5.2. *The Jacobian determinant at P_+ evaluated in [1] is $5s_0(1 - c_0)^2/Q_0 \approx 2.245$, which is numerically close to but not exactly $\sqrt{5} \approx 2.236$. The exact value involves $(1 - c_0) = \varphi^{-1}$, $Q_0 = \varphi^{-1/2}$, and $s_0 = \sqrt{(3\sqrt{5} - 5)/2}$. Lemma 5.1 identifies $\sqrt{5}$ as the rate of vanishing of the chosen monic polynomial $\mathcal{P}$. It is not an invariant of the zero germ by itself: replacing $\mathcal{P}$ by $a\mathcal{P}$, $a \neq 0$, preserves the zero set but changes the derivative to $|a|\sqrt{5}$. The same constant reappears as an intensity, rather than a rate: in the companion paper [2], the Stokes caustic on the Poincaré sphere induced by the fold locus of $\mathcal{S}$ has total intensity exactly $\sqrt{5}/2 = |\mathcal{P}'(c_0)|/2$ at its far endpoint—an independent derivation converging on the same constant. This repetition is a noteworthy structural observation, but it does not by itself define capacity.*

Theorem 5.3 (Intrinsic Sublevel Exponent). *For the Whitney fold $f_f(x, y) = (x, y^2)$ with squared residual $K_f = \|f_f\|^2 = x^2 + y^4$,*

$$V_{K_f}(\varepsilon) = B\left(\frac{1}{4}, \frac{3}{2}\right) \varepsilon^{3/4}.$$

For the Whitney cross-cap $f_\times(x, y) = (x, xy, y^2)$ with $K_\times = \|f_\times\|^2 = x^2(1 + y^2) + y^4$,

$$V_{K_\times}(\varepsilon) = B\left(\frac{1}{4}, \frac{3}{2}\right) \varepsilon^{3/4} (1 + o(1)).$$

Thus both local models have the intrinsic sublevel, or singular-learning, exponent $\lambda = 3/4$.

Proof. For the fold,

$$\begin{aligned} V_{K_f}(\varepsilon) &= 4 \int_0^{\varepsilon^{1/4}} \sqrt{\varepsilon - y^4} dy \\ &= 4\varepsilon^{3/4} \int_0^1 \sqrt{1 - t^4} dt = B\left(\frac{1}{4}, \frac{3}{2}\right) \varepsilon^{3/4}. \end{aligned}$$

For the cross-cap, set $x = \varepsilon^{1/2}s$ and $y = \varepsilon^{1/4}t$. The Jacobian is $\varepsilon^{3/4}$ and the rescaled sublevel condition is $s^2(1 + \varepsilon^{1/2}t^2) + t^4 \leq 1$, which converges to $s^2 + t^4 \leq 1$. Dominated convergence gives the stated constant and exponent. This agrees with the metric interpretation of real log-canonical thresholds by sublevel-volume asymptotics [6]. □

Remark 5.4 (A Logarithmic Signal-Resolution Observation). *If an operational model independently identifies $S = \varepsilon^{-1}$, then Theorem 5.3 gives*

$$\mathcal{C}_K(S^{-1}) = \frac{3}{4} \log S - \log B\left(\frac{1}{4}, \frac{3}{2}\right) + o(1).$$

It is worth noting that a logarithmic law survives, but its intrinsic local coefficient is $3/4$, not $\sqrt{5}$. Calling this quantity a Shannon capacity would still require a channel law and a noise/resource constraint; no such physical identification is asserted here.

Proposition 5.5 (Residual Geometry Does Not Fix Channel Capacity). *The residual germ and its sublevel volumes do not uniquely determine a Shannon capacity.*

Proof. On fixed binary input and output alphabets, the constant channel $P(Y = 0 | X = 0) = P(Y = 0 | X = 1) = 1$ has capacity zero, whereas the noiseless channel $P(Y = X | X) = 1$ has capacity $\log 2$. The source and observation labels may be held fixed while the stochastic kernel changes. Since the residual germ supplies no kernel, input constraint, or noise scale, it cannot select between these values. $\square$

5.1 Complex Analytic Completion of the Cross-Cap Residual

The following completion is made only after extending the coefficients of $K_\times$ from $\mathbb{R}$ to $\mathbb{C}$. This is explicit algebraic input, not a reconstruction from the real image topology.

Theorem 5.6 (A_3 residual germ). *The complex analytic germ*

$$K_\times(x, y) = x^2(1 + y^2) + y^4$$

is right-equivalent at the origin to $f(X, Y) = X^2 + Y^4$. Its Milnor and Tjurina numbers are $\mu(K_\times) = \tau(K_\times) = 3$.

Proof. On a small disk about $y = 0$, choose the holomorphic branch $u(y) = (1 + y^2)^{1/2}$ with $u(0) = 1$, and set $X = xu(y)$, $Y = y$. The Jacobian determinant at the origin is one, so this is a local biholomorphism, and $K_\times = X^2 + Y^4$. Its Jacobian algebra is

$$\frac{\mathbb{C}\{X, Y\}}{(2X, 4Y^3)} \cong \frac{\mathbb{C}\{Y\}}{(Y^3)},$$

with basis $1, Y, Y^2$. Weighted homogeneity gives $f = \frac{1}{2}X\partial_X f + \frac{1}{4}Y\partial_Y f$, so f lies in the Jacobian ideal and the Tjurina algebra has the same dimension. $\square$

Theorem 5.7 (Milnor fiber and finite monodromy). *The global Milnor fiber*

$$U = \{(X, Y) \in \mathbb{C}^2 : X^2 + Y^4 = 1\}$$

is a genus-one curve with two points removed. Hence $\dim H^1(U; \mathbb{Q}) = 3$. Its geometric monodromy can be chosen as

$$h(X, Y) = (-X, iY), \quad h^4 = \text{id},$$

and on first cohomology

$$\text{spec}(h^*) = \{-i, 1, i\}, \quad \det(tI - h^*) = (t - 1)(t^2 + 1).$$

Proof. Weighted homogeneity identifies U with the local Milnor fiber [7, 8]. Compactify it as

$$E : X^2 + Y^4 = Z^4 \subset \mathbb{P}(2, 1, 1).$$

Projection to $[Y : Z] \in \mathbb{P}^1$ is a double cover branched at four simple points. Riemann–Hurwitz gives $g(E) = 1$. The fiber above $[Y : Z] = [1 : 0]$ contains two unramified points, so $U = E \setminus D$ with $|D| = 2$ and $b_1(U) = 2g(E) + |D| - 1 = 3$.

The displayed h is the weighted-homogeneous monodromy. The differential $\omega = dY/X$ spans $H^{1,0}(E)$ and satisfies $h^*\omega = -i\omega$; its conjugate has eigenvalue i . With $s = 1/Y$ and $W = X/Y^2$ near infinity, $h(s, W) = (-is, W)$, so both points of D and the logarithmic boundary class are fixed. These three classes exhaust $H^1(U; \mathbb{C})$. $\square$

5.2 Figure-Eight Links and the Weighted Phase Lift

The real cross-cap and its chosen A_3 completion carry two related, but distinct, figure-eight constructions. The first is the link of the real cross-cap image. The second is a planar shadow of the weighted phase orbit in $\mathbb{C}^2$. Keeping them separate prevents a projected self-intersection from being mistaken for a singularity of the lifted source.

Proposition 5.8 (The cross-cap link is a figure-eight quotient). *For $r > 0$, let*

$$L_r := \{(x, y) \in \mathbb{R}^2 : K_\times(x, y) = r^2\}.$$

Then L_r is a smooth circle. The restriction $f_\times|_{L_r} : L_r \rightarrow S_r^2$ is an immersion with exactly one transverse double point,

$$f_\times(0, \sqrt{r}) = f_\times(0, -\sqrt{r}) = (0, 0, r).$$

Consequently its image is homeomorphic to the figure-eight graph $S^1 \vee S^1$. Moreover,

$$\tilde{f}_\times(x, y) := (x, xy, y^2, y) \in \mathbb{R}^4$$

is an embedding and satisfies the reconstruction identity

$$\pi_{123} \circ \tilde{f}_\times = f_\times,$$

where π_{123} forgets the last coordinate.

Proof. The gradient

$$\nabla K_\times = (2x(1 + y^2), 2x^2y + 4y^3)$$

vanishes only at the origin, which is not in L_r . Thus L_r is a smooth compact one-manifold. It is connected because it is the union of the two graphs

$$x = \pm \sqrt{\frac{r^2 - y^4}{1 + y^2}}, \quad -\sqrt{r} \leq y \leq \sqrt{r},$$

joined at their two endpoints; hence $L_r \cong S^1$.

If $f_\times(x, y) = f_\times(x', y')$, then $x = x'$ and $y'^2 = y^2$. The case $y' = y$ gives the same source point. If $y' = -y$, equality of the second coordinates gives $xy = -xy$. The only distinct pair on L_r is therefore $(0, \sqrt{r})$ and $(0, -\sqrt{r})$. At these points the source tangent is spanned by $(1, 0)$, and its two image tangents are

$$(1, \sqrt{r}, 0), \quad (1, -\sqrt{r}, 0).$$

Their cross product is $(0, 0, -2\sqrt{r}) \neq 0$, so the double point is transverse. Identifying exactly two distinct points of a circle gives $S^1 \vee S^1$. Finally, the first and fourth coordinates of $f_\times$ recover (x, y) , so it is an embedding, and the displayed projection identity is immediate. $\square$

Theorem 5.9 (A_3 weighted phase orbit and figure-eight shadow). *The primitive positive integral weights of $X^2 + Y^4$ are $(2, 1)$. Define*

$$\Phi_s(X, Y) := (e^{2is}X, e^{is}Y).$$

Then

$$f(\Phi_s(X, Y)) = e^{4is} f(X, Y), \quad \Phi_{\pi/2} = h.$$

For any $a, b > 0$ satisfying $a^2 + b^4 = 1$, the orbit

$$\Gamma_{a,b}(s) := (ae^{2is}, be^{is}), \quad s \in \mathbb{R}/2\pi\mathbb{Z},$$

is an embedded circle in $\mathbb{C}^2$. Its rectangular projection

$$P(\Gamma_{a,b}(s)) = (\operatorname{Im} Y, \operatorname{Im} X) = (b \sin s, a \sin 2s)$$

has exactly one transverse double point and obeys

$$v^2 = 4a^2 \frac{u^2}{b^2} \left(1 - \frac{u^2}{b^2} \right).$$

Adding the single lost coordinate $\operatorname{Re} Y = b \cos s$ reconstructs the lifted orbit. Along the phase parameter,

$$\frac{d^2 X}{ds^2} + 4X = 0, \quad \frac{d^2 Y}{ds^2} + Y = 0,$$

so the dimensionless frequency ratio is exactly $2 : 1$.

Proof. Weighted homogeneity requires $2w_X = 4w_Y$; primitivity therefore gives $(w_X, w_Y) = (2, 1)$. Direct substitution proves the covariance formula, and $s = \pi/2$ gives $(-X, iY) = h(X, Y)$. Since $b > 0$, the Y coordinate determines s modulo 2π , so $\Gamma_{a,b}$ is embedded.

Set $u = b \sin s$ and $v = a \sin 2s$. Eliminating s gives Equation (5.9). If two parameter values have the same (u, v) , the only distinct pair is $s = 0, \pi$. Their projected tangents are $(b, 2a)$ and $(-b, 2a)$, whose determinant is $4ab \neq 0$. The pair $(\operatorname{Re} Y, \operatorname{Im} Y)$ recovers be^{is} and hence s , after which $X = ae^{2is}$ is fixed. The oscillator equations follow by differentiating Equation (5.9) twice. $\square$

Proposition 5.10 (The Euclidean phase action does not fix the bridge). *For the orbit family in Theorem 5.9, the Euclidean quadratic action is*

$$\mathcal{E}(a, b) := \int_0^{2\pi} \|\Gamma'_{a,b}(s)\|^2 ds = 2\pi(4a^2 + b^2).$$

Subject to $a^2 + b^4 = 1$, its minimum is the degenerate orbit $(a, b) = (0, 1)$. Its unique interior critical point satisfies

$$b^2 = \frac{1}{8}, \quad a^2 = \frac{63}{64},$$

and is a strict maximum. Thus this action fixes neither a non-degenerate figure-eight amplitude nor an absolute clock.

Proof. Put $z = b^2 \in [0, 1]$. The constraint gives $a^2 = 1 - z^2$, and hence

$$\frac{\mathcal{E}}{2\pi} = 4 + z - 4z^2.$$

This polynomial is strictly concave. Its minimum occurs at the endpoint $z = 1$, where $a = 0$, while its only interior critical point $z = 1/8$ is the strict maximum. Replacing s by $\omega_0 t$ leaves the ratio $2 : 1$ unchanged for every $\omega_0 > 0$, so the geometry supplies no absolute time scale. $\square$

Remark 5.11 (Wave and reconstruction scope). *The phase orbit gives an exact two-frequency oscillator, not yet a spacetime wave equation. Treating its planar shadow as a metric graph would require an independent matching condition at the four-valent vertex [9]. The normalization lift instead keeps the two source points distinct and therefore specifies source continuation without creating a physical scattering vertex. The value $b^4 = 1/64$ at the interior maximum of Proposition 5.10 is metric- and action-dependent; it is not identified with $1/(64\pi^2)$.*

Proposition 5.12 (The actual ruled cross-caps remain on the cusp boundary). *Let $P_\sigma = (\sigma t_0, u_0)$, $\sigma \in \{+1, -1\}$, be the two lateral cross-caps of the ruled surface S , and write its unique oriented Euclidean Whitney normal form as in [10, Proposition 2.1]. Its second-order invariants $(a_\sigma, b_\sigma, c_\sigma)$ are*

$$\begin{aligned} a_\sigma &= -\frac{M^2}{2N^{3/2}}, & b_\sigma &= \sigma \frac{\sqrt{M^2 R}}{N}, & c_\sigma &= \frac{\sqrt{2R}}{N^{1/4}}, \\ N &= \frac{-335 + 162\sqrt{5}}{12}, & M^2 &= \frac{-520 + 249\sqrt{5}}{27}, & R &= \frac{20(140 + 11\sqrt{5})}{11397}. \end{aligned}$$

In particular,

$$(a_+, b_+, c_+) \approx (-0.1991214663, 0.2763032014, 0.6191941949),$$

while $a_- = a_+$, $b_- = -b_+$, and $c_- = c_+$. The orientation-free data are a_σ , $|b_\sigma|$, and c_σ . They obey the exact identity

$$b_\sigma^2 + a_\sigma c_\sigma^2 = 0. \quad (2)$$

Consequently the polynomial selecting returns of the principal-curvature exceptional trace to the origin factors as

$$c_\sigma \tau^2 + 2b_\sigma \tau - a_\sigma c_\sigma = c_\sigma \left(\tau + \frac{b_\sigma}{c_\sigma} \right)^2.$$

The trace therefore has a double return direction and a cusp, not a transverse node or figure-eight.

Proof. Put $c_0 = (3 - \sqrt{5})/2$, $u_0 = (\sqrt{5} - 1)/4$, and $s_\sigma = \sigma\sqrt{2 - 3c_0}$. At P_σ choose the adapted pair

$$\xi_0 = \frac{1}{\sqrt{3}}\partial_u, \quad \eta_0 = -\partial_t + \lambda_\sigma \partial_u, \quad \lambda_\sigma = -\frac{c_0 s_\sigma}{2(1 - c_0)}.$$

Let Π_N be orthogonal projection onto the normal plane of the rank-one image and set

$$A = \Pi_N(S_{tt} - 2\lambda_\sigma S_{tu}), \quad B = \Pi_N(-S_{tu}/\sqrt{3}).$$

Direct substitution into the ruled-surface derivatives gives

$$\|A\|^2 = N, \quad \langle A, B \rangle^2 = M^2, \quad \|B\|^2 - \frac{\langle A, B \rangle^2}{N} = R.$$

The oriented Gram–Schmidt reduction to the unique Whitney normal form gives the displayed $(a_\sigma, b_\sigma, c_\sigma)$. Substitution then yields

$$b_\sigma^2 = \frac{M^2 R}{N^2} = -a_\sigma c_\sigma^2,$$

which proves (2). By the pedal formula for the principal-curvature trace [10, Equation (2.8)], its nonzero scalar factor vanishes precisely at the roots of the displayed quadratic. Its discriminant is four times $b_\sigma^2 + a_\sigma c_\sigma^2$, hence zero. $\square$

Remark 5.13 (Principal-curvature bridge no-go). *This conclusion uses only the actual second jet and no fitted scale. It is also structural: S is affine in its ruling coordinate, so $S_{uu} = 0$, and the canonical normal-form shear forces (2). Higher jets may change the geometry away from the exceptional trace, but cannot change this two-jet discriminant. Thus the principal-curvature construction supplies neither the amplitudes (a, b) of Theorem 5.9 nor a canonical identification with Proposition 5.8.*

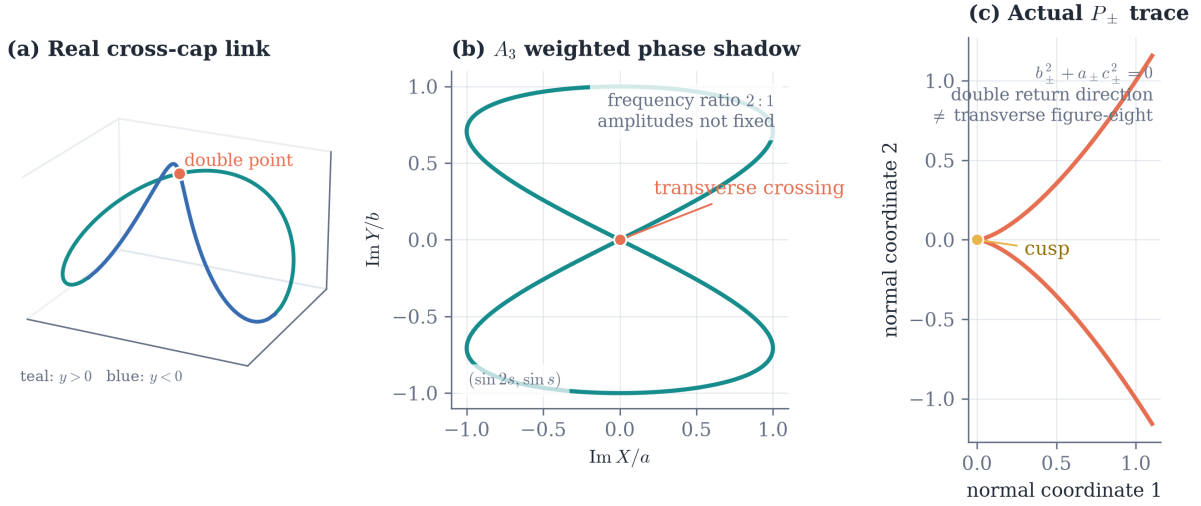


Figure 4: Three distinct projected structures. (a) The positive-radius link of the real cross-cap has a transverse double point in its image (Proposition 5.8). (b) The chosen A_3 weighted phase orbit has a separate 2 : 1 Lissajous shadow (Theorem 5.9); its amplitudes remain unfixed. (c) The actual ruled-surface germs at $P_{\pm}$ instead have a cusp, because $b_{\pm}^2 + a_{\pm} c_{\pm}^2 = 0$ (Proposition 5.12).

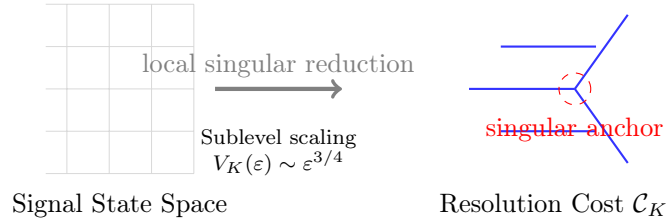


Figure 5: Schematic of the intrinsic sublevel law in Theorem 5.3. It records a geometric resolution cost, not a communication-channel or holographic capacity.

6 Immersion Singularities and the Golden Ratio

The topological stability of the realized signal manifold is governed by immersion singularities—points where $\mathcal{P}(\mathcal{S})$ fails to be a local immersion.

Proposition 6.1 (Governing Invariant). *The immersion singularities of $\mathcal{S}$ are governed by $c^2 - 3c + 1 = 0$, whose unique root in $(0, 1)$ is $c_0 = \varphi^{-2} = (3 - \sqrt{5})/2$.*

Proof. This is Theorem 1.3 of [1], established there by solving $V_z = V_y = 0$ for the components of the cross product $V = S_t \times S_u$: $V_z = 0$ gives $u = (1 - c)/2$, $V_y = 0$ gives $u = c(2 - c)/2$, and equating the two yields $c^2 - 3c + 1 = 0$; vanishing of V_x at the resulting point is verified there by direct substitution. We record only the statement needed below and omit the derivation. $\square$

Proposition 6.2 (Cross-Cap Classification). *At the three rank-one points of the smooth extension of (1), namely the central point $P_0 = (0, 1)$ and the lateral points $P_{\pm}$, the determinant*

$$\Delta := \det(S_u, S_{ut}, S_{tt})$$

is non-zero. More precisely,

$$\Delta(P_0) = 1, \quad \Delta(P_+) = \Delta(P_-) = -5\sqrt{\sqrt{5} - 2}.$$

Consequently each smooth interior germ is a Whitney cross-cap. At P_0 the original parameter rectangle shows only the corresponding boundary half-germ.

Proof. At a rank-one point write $S_t = aS_u$ and use the kernel direction $\partial_t - a\partial_u$. Since (1) is affine in u , the standard cross-cap determinant in these adapted coordinates reduces to $\det(S_u, S_{ut}, S_{tt})$ [5, 4]. Direct differentiation gives $\Delta(P_0) = 1$; at the lateral points, reduction by $c_0^2 = 3c_0 - 1$ gives $\Delta(P_{\pm}) = -5\sqrt{\sqrt{5} - 2} \neq 0$. $\square$

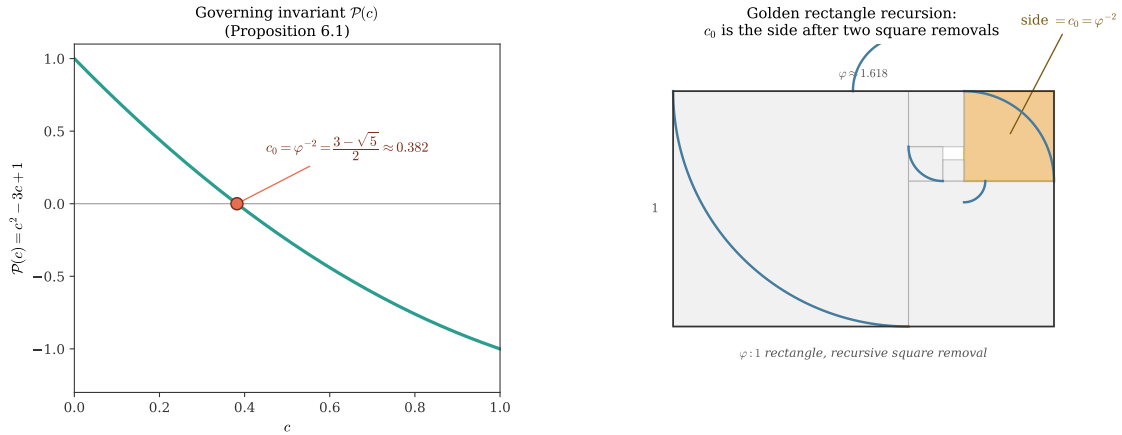


Figure 6: Left: the governing invariant $\mathcal{P}(c) = c^2 - 3c + 1$ (Proposition 6.1) with its unique root $c_0 = \varphi^{-2} \approx 0.382$ in $(0, 1)$. Right: the classical golden rectangle recursion (a $\varphi:1$ rectangle with unit squares recursively removed). Since each removal shrinks the residual rectangle by a linear factor φ^{-1} , the side of the *second* residual square (amber) is exactly $\varphi^{-2} = c_0$: the same constant that governs the immersion singularities of $\mathcal{S}$ arises as a length in the classical golden-ratio construction, independently of the surface.

7 Spectral Stability and Near-Orthogonality

7.1 Formal Self-Adjointness

Theorem 7.1 (Formal Self-Adjointness). *On the domain $\mathcal{D} = C_c^\infty((1, \infty))$ in $L^2([1, \infty), dx)$, the formal adjoint of $\mathcal{T}_\sigma = -i(x\partial_x + \sigma)$ is*

$$\mathcal{T}_\sigma^* = -i\left(x \frac{d}{dx} + (1 - \sigma)\right) = \mathcal{T}_{1-\sigma}.$$

Consequently $\mathcal{T}_\sigma$ is formally self-adjoint ($\mathcal{T}_\sigma = \mathcal{T}_\sigma^*$) if and only if $\sigma = \frac{1}{2}$.

Proof. Let $f, g \in \mathcal{D}$. All boundary terms at $x = 1$ vanish because f, g are compactly supported in $(1, \infty)$. Compute:

$$\langle \mathcal{T}_\sigma f, g \rangle = \int_1^\infty \overline{-i(xf'(x) + \sigma f(x))} g(x) dx = i \int_1^\infty (x\bar{f}'g + \sigma\bar{f}g) dx.$$

Integrate by parts (boundary term $[x\bar{f}g]_1^\infty = 0$):

$$\int_1^\infty x\bar{f}'g dx = - \int_1^\infty \bar{f}(g + xg') dx.$$

Substituting:

$$\begin{aligned} \langle \mathcal{T}_\sigma f, g \rangle &= i \left(- \int_1^\infty \bar{f}g dx - \int_1^\infty \bar{f}(xg') dx + \sigma \int_1^\infty \bar{f}g dx \right) \\ &= \int_1^\infty \overline{f(x)} [-i(xg'(x) + (1 - \sigma)g(x))] dx = \langle f, \mathcal{T}_{1-\sigma} g \rangle. \end{aligned}$$

Hence $\mathcal{T}_\sigma^* = \mathcal{T}_{1-\sigma}$, and $\mathcal{T}_\sigma = \mathcal{T}_\sigma^*$ iff $\sigma = 1 - \sigma$, i.e. $\sigma = \frac{1}{2}$. $\square$

Remark 7.2 (Von Neumann Deficiency Indices). *On $L^2([1, \infty))$ with domain $C_c^\infty((1, \infty))$, the von Neumann deficiency indices of $\mathcal{T}_{1/2}$ are $(n_+, n_-) = (1, 0)$: the solution $\psi_+(x) = x^{-3/2}$ to $(\mathcal{T}_{1/2})^*\psi = i\psi$ lies in $L^2([1, \infty))$ (since $\int_1^\infty x^{-3} dx = \frac{1}{2} < \infty$), while the solution $\psi_-(x) = x^{1/2}$ to $(\mathcal{T}_{1/2})^*\psi = -i\psi$ does not ($\int_1^\infty x dx = \infty$). Unequal deficiency indices $(1, 0)$ mean $\mathcal{T}_{1/2}$ on this domain is not essentially self-adjoint and admits no self-adjoint extension.*

By contrast, on $L^2((0, \infty))$ with domain $C_c^\infty((0, \infty))$, neither $x^{-3/2}$ (diverges at 0) nor $x^{1/2}$ (diverges at ∞) lies in the space, so deficiency indices are $(0, 0)$ and $\mathcal{T}_{1/2}$ is essentially self-adjoint there—as confirmed by the Mellin–Plancherel isomorphism $\mathcal{M} : L^2((0, \infty)) \rightarrow L^2(\mathbb{R})$ which conjugates $\mathcal{T}_{1/2}$ to multiplication by ω [16]. For the signal-theoretic setting with the semi-infinite domain $[1, \infty)$, Theorem 7.1 (formal self-adjointness) is the appropriate statement.

Corollary 7.3 (Real Phase Frequencies). *The formal eigenfunctions $f_\omega(x) = x^{-(\sigma+i\omega)}$, $\omega \in \mathbb{R}$, satisfy $\mathcal{T}_\sigma f_\omega = -\omega f_\omega$ with eigenvalue $-\omega \in \mathbb{R}$ for all σ . These are generalized eigenfunctions rather than vectors in the half-line L^2 space, so no Hilbert-space orthogonality follows from formal self-adjointness alone.*

Proof. Direct computation:

$$\begin{aligned}
\mathcal{T}_\sigma [x^{-(\sigma+i\omega)}] &= -i(x \cdot (-(\sigma + i\omega)) x^{-(\sigma+i\omega)-1} + \sigma x^{-(\sigma+i\omega)}) \\
&= -i(-(\sigma + i\omega) + \sigma) x^{-(\sigma+i\omega)} \\
&= -i(-i\omega) x^{-(\sigma+i\omega)} \\
&= -\omega x^{-(\sigma+i\omega)}.
\end{aligned}$$

On the full multiplicative line $(0, \infty)$, their distributional orthogonality is instead supplied by the Mellin–Plancherel theorem. The half-line realization has the deficiency obstruction recorded in Remark 7.2. $\square$

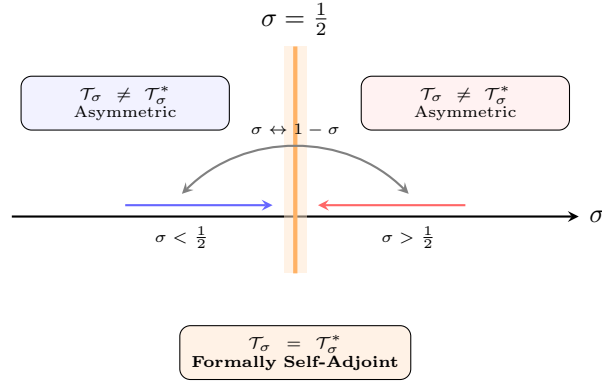


Figure 7: Formal self-adjointness of $\mathcal{T}_\sigma$. At $\sigma = \frac{1}{2}$, $\mathcal{T}_\sigma^* = \mathcal{T}_{1-\sigma} = \mathcal{T}_\sigma$ at the level of the compactly supported formal domain. Away from $\sigma = \frac{1}{2}$, the formal adjoint is a different operator; the separate generalized-eigenvalue calculation remains real for every σ .

7.2 Uniform L^2 Near-Orthogonality

To quantify the stability of interference nodes, we establish a near-orthogonality theorem for the phasor family $\{\psi_n(t) = e^{-it \ln n}\}$.

Theorem 7.4 (Gram Matrix Bound). *For a finite set $\mathbf{S} \subset \mathbb{N}$ with $|\mathbf{S}| = M$, let $G = (\langle \psi_n, \psi_m \rangle_T)$ be the Gram matrix with entries $\langle \psi_n, \psi_m \rangle_T = \frac{1}{T} \int_0^T \psi_n \overline{\psi_m} dt$. If $\delta := \min_{n \neq m \in \mathbf{S}} |\ln(n/m)| > 0$, then*

$$\|G - I\|_{\text{op}} \leq \frac{2(M-1)}{T\delta}.$$

Proof. The off-diagonal entry satisfies $|\langle \psi_n, \psi_m \rangle_T| = \frac{1}{T} \left| \int_0^T e^{-it \ln(n/m)} dt \right| \leq \frac{2}{T|\ln(n/m)|} \leq \frac{2}{T\delta}$. For each row n , the sum of off-diagonal moduli is at most $(M-1) \cdot 2/(T\delta)$. By Gershgorin's circle theorem [26], every eigenvalue of $G - I$ has modulus at most $(M-1) \cdot 2/(T\delta)$, giving the operator-norm bound. $\square$

Remark 7.5 (Asymptotic Orthogonality). *The bound $\|G - I\|_{\text{op}} = O(T^{-1})$ as $T \rightarrow \infty$ confirms that the phasors become asymptotically orthogonal, with the rate controlled by the minimal log-separation δ of the node set.*

Proposition 7.6 (Linear Independence and Positive Definiteness). *Under the hypotheses of Theorem 7.4, the phasor family $\{\psi_n\}_{n \in \mathbf{S}}$ is linearly independent in $L^2([0, T])$ whenever*

$$T > \frac{2(M-1)}{\delta}.$$

Equivalently, the Gram matrix G is positive definite, with $\lambda_{\min}(G) \geq 1 - 2(M-1)/(T\delta) > 0$.

Proof. From Theorem 7.4, $\|G - I\|_{\text{op}} \leq 2(M-1)/(T\delta)$. Weyl's eigenvalue interlacing gives $\lambda_{\min}(G) \geq \lambda_{\min}(I) - \|G - I\|_{\text{op}} = 1 - 2(M-1)/(T\delta)$. When $T > 2(M-1)/\delta$ this is strictly positive, so $G \succ 0$ and the phasors are linearly independent. $\square$

Remark 7.7 (Bohr Interpretation). *The limit $G \rightarrow I$ has a precise compact-group realization through the prime-exponent characters of the Bohr torus; see Theorem 8.4. No freeness assertion of rank M is needed, and multiplicative relations among the selected integers are retained.*

Corollary 7.8 (Condition Number and Stability Bound). *Under the hypotheses of Theorem 7.4 with $\varepsilon := 2(M-1)/(T\delta) < 1$, the Gram matrix satisfies*

$$1 - \varepsilon \leq \lambda_{\min}(G) \leq \lambda_{\max}(G) \leq 1 + \varepsilon,$$

and the condition number is bounded by

$$\kappa(G) := \frac{\lambda_{\max}(G)}{\lambda_{\min}(G)} \leq \frac{1 + \varepsilon}{1 - \varepsilon}.$$

Moreover, the log-determinant satisfies $\log \det(G) \geq M \log(1 - \varepsilon)$, so the information-geometric volume of the phasor family is at most $M\varepsilon/(1 - \varepsilon)$ below the ideal orthogonal value.

Proof. Both eigenvalue bounds follow from Theorem 7.4 via the Weyl perturbation inequality: $|\lambda_k(G) - \lambda_k(I)| \leq \|G - I\|_{\text{op}}$. Since $\lambda_k(I) = 1$ for all k , we get $1 - \varepsilon \leq \lambda_k(G) \leq 1 + \varepsilon$. The condition number bound is immediate. The log-determinant bound follows from $\det(G) = \prod_k \lambda_k(G) \geq (1 - \varepsilon)^M$, taking logarithms and using $\log(1 - \varepsilon) \geq -\varepsilon/(1 - \varepsilon)$ for $0 < \varepsilon < 1$. $\square$

8 Independent Structural Completions and Scope Limits

This section records the independent structures that can be derived without a channel law, a particle-mass input, or an arithmetic-geometric identification. Its role is both constructive and negative: it supplies canonical replacements where they exist and states the remaining obstructions where they do not.

8.1 Mellin Half-Density and the Dirichlet–Hardy Boundary

Let $G = (\mathbb{R}_{>0}, \times)$ with Haar measure dt/t , and let $A = -it\partial_t$ be the self-adjoint Mellin generator on the full group.

Proposition 8.1 (Full-Line Plancherel Realization). *The map*

$$W : L^2(\mathbb{R}_{>0}, dt/t) \longrightarrow L^2(\mathbb{R}_{>0}, dt), \quad (Wf)(t) = t^{-1/2}f(t),$$

Finite-time spectral stability of a phasor family

Illustrative family $S = \{2, 3, 5, 7, 11\}$, $T = 80$, $\delta = 0.336$; $\varepsilon = 2(M-1)/(T\delta) \approx 0.297$.

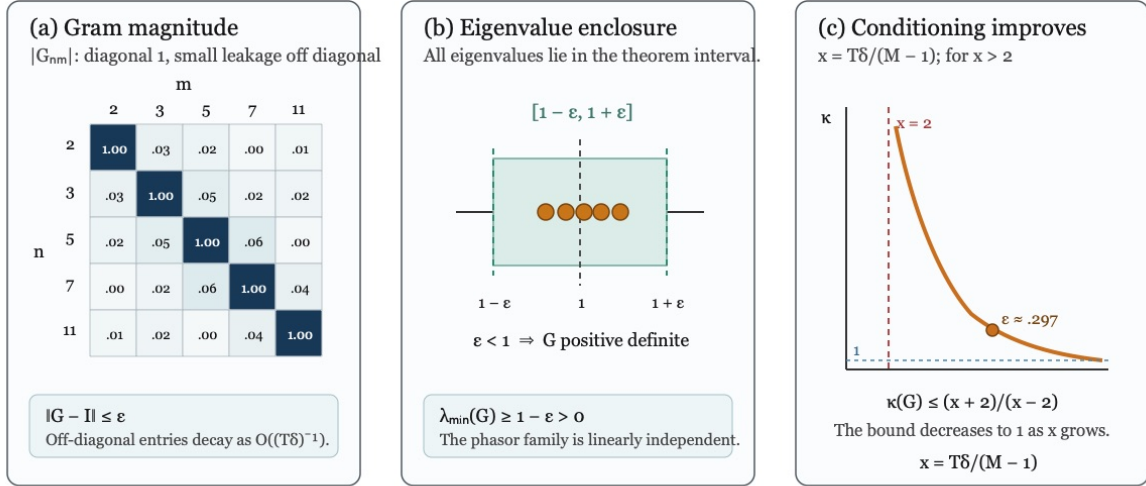


Figure 8: Finite-time spectral stability of a phasor family. Panel (a) shows one illustrative Gram matrix; panels (b) and (c) summarize the exact enclosure and condition-number bounds of Theorem 7.4, Proposition 7.6, and Corollary 7.8. With $x = T\delta/(M-1) > 2$, the displayed condition bound is the equivalent form $\kappa(G) \leq (x+2)/(x-2)$. The numerical entries illustrate the inequalities and are not additional data or assumptions.

is unitary and satisfies

$$WAW^{-1} = -i \left(t\partial_t + \frac{1}{2} \right) = \mathcal{T}_{1/2}.$$

Thus the half-density $t^{-1/2}$ is fixed by the change from multiplicative Haar measure to Lebesgue measure. On $[1, \infty)$ this operator is only the half-line restriction and retains the deficiency obstruction of Remark 7.2; Plancherel does not make that restriction self-adjoint.

Proof. Unitarity follows from $\int_0^\infty |t^{-1/2}f(t)|^2 dt = \int_0^\infty |f(t)|^2 dt/t$. For a smooth compactly supported g ,

$$WAW^{-1}g = t^{-1/2}[-it\partial_t(t^{1/2}g)] = -i(tg' + \frac{1}{2}g).$$

The full-line self-adjoint statement is the Mellin–Plancherel theorem [16]. $\square$

Theorem 8.2 (Dirichlet–Hardy Evaluation Boundary). *Let*

$$\mathcal{H}_D^2 = \left\{ f(s) = \sum_{n \geq 1} a_n n^{-s} : \sum_{n \geq 1} |a_n|^2 < \infty \right\}.$$

For $\sigma = \Re s > 1/2$, point evaluation is bounded and

$$|f(s)| \leq \|(a_n)\|_{\ell^2} \zeta(2\sigma)^{1/2}.$$

For $\sigma \leq 1/2$ the evaluation vector $(n^{-\bar{s}})_n$ is not in ℓ^2 , so point evaluation is not a bounded functional on $\mathcal{H}_D^2$. Hence $\sigma = 1/2$ is the canonical analytic boundary of this Hilbert space of Dirichlet series.

Proof. The bound is Cauchy–Schwarz and $\sum_{n \geq 1} |n^{-s}|^2 = \zeta(2\sigma)$. Conversely, bounded evaluation on the coefficient Hilbert space would, by the Riesz representation theorem, require $(n^{-\bar{s}})_n \in \ell^2$, which fails exactly when $\sum n^{-2\sigma}$ diverges [17]. $\square$

Remark 8.3 (A Noteworthy Half-Density Observation). *The same value $\sigma = 1/2$ occurs in two independent analytic roles: formal-adjoint symmetry on the half-line and the evaluation boundary of $\mathcal{H}_D^2$. This coincidence is worth recording for signal analysis, but it is not a Deligne weight formula or a proof of a physical critical line.*

8.2 Bohr–Haar Completion of the Phasor Gram Limit

For $n = \prod_p p^{v_p(n)}$, let $\alpha(n) = (v_p(n))_p \in \mathbb{Z}^{(\mathcal{P})}$ and define the character $\chi_n(z) = \prod_p z_p^{v_p(n)}$ on the infinite torus $\mathbb{T}^{\mathcal{P}}$.

Theorem 8.4 (Bohr–Haar Orthogonality). *With normalized Haar measure m_H ,*

$$\int_{\mathbb{T}^{\mathcal{P}}} \chi_n(z) \overline{\chi_m(z)} dm_H(z) = \delta_{nm}.$$

Moreover,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T n^{-it} m^{it} dt = \delta_{nm}.$$

Thus the limiting identity Gram matrix is the character orthogonality relation on the Bohr torus, while Theorem 7.4 quantifies its finite-time approximation.

Proof. Distinct exponent vectors define distinct torus characters by unique prime factorization, and non-trivial characters have zero Haar integral. For $n \neq m$, direct integration gives

$$\left| \frac{1}{T} \int_0^T e^{-it \log(n/m)} dt \right| \leq \frac{2}{T |\log(n/m)|} \rightarrow 0;$$

the diagonal integral equals one. $\square$

Remark 8.5 (What the Arithmetic Structure Does Not Fix). *The multiplicative group generated by a finite set of M integers need not have rank M ; for example, $\{2, 4\}$ generates a rank-one group. The canonical completion is therefore the torus determined by the prime-exponent lattice, not a freely declared $\mathbb{Z}^M$ and not a Weil-cohomological object. The resemblance between phase decorrelation and arithmetic character orthogonality remains a useful observation point only.*

8.3 The Constructible Defect of a Cross-Cap

Let $\nu : \tilde{X} \rightarrow X$ be the local normalization of a cross-cap image and let $j : L \hookrightarrow X$ denote its locally closed open double stratum, obtained by removing the pinch point from the double locus.

Proposition 8.6 (Normalization Defect Sheaf). *For rational constant sheaves, the quotient*

$$\mathcal{Q}_\nu := \nu_* \mathbb{Q}_{\tilde{X}} / \mathbb{Q}_X$$

is canonical and is supported on L . More precisely, $\mathcal{Q}_\nu \simeq j_! \mathcal{L}_{\text{sgn}}$, where $\mathcal{L}_{\text{sgn}}$ is the rank-one anti-invariant local system of the two-sheet cover. Its stalk is 0 where ν has one preimage and $\mathbb{Q}$ where ν has two preimages. On a simply connected local branch, choosing an order of the sheets trivializes $\mathcal{L}_{\text{sgn}} \cong \mathbb{Q}_L$.

Proof. At a one-sheet point, the stalk map $\mathbb{Q} \rightarrow \mathbb{Q}$ is an isomorphism. At a two-sheet point it is the diagonal map $\mathbb{Q} \rightarrow \mathbb{Q}^2$, whose cokernel is canonically the anti-invariant one-dimensional quotient. Sheet exchange acts on that quotient by -1 . At the pinch point the two sheets meet and the quotient stalk returns to zero. These stalk maps commute with restriction, giving extension by zero of the sign local system from L ; this is the elementary constructible-sheaf description of the normalization defect [18, 19]. $\square$

Remark 8.7 (Scope of the Condensed View). *A compact Hausdorff realization does define a condensed set, but that fact alone supplies neither a canonical structure sheaf nor a tangent module whose freeness detects the singularity. Proposition 8.6 is the canonical local invariant available from the normalization map. No stronger condensed-smoothness or exceptional-pullback identity is asserted. The obstruction is elementary: the same one-point topological space supports, for example, the nonisomorphic stalk rings $\mathbb{Q}$ and $\mathbb{Q}[\epsilon]/(\epsilon^2)$. Once the polynomial complexification is chosen, however,*

$$U_{\mathbb{Q}} = \text{Spec } \mathbb{Q}[X, Y]/(X^2 + Y^4 - 1)$$

has its ordinary algebraic structure sheaf. That sheaf is constructed from the stated coordinate ring; it is not recovered from an underlying condensed set alone [14].

8.4 The Polar Fold of the Observation Field

Proposition 8.8 (Polar Whitney Fold). *At the boundary point $(t, u) = (0, 1)$, the planar map $\Psi = (N, M)$ has*

$$D\Psi(0, 1) = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $\tau = t$ and $\rho = 1 - u$,

$$N(\tau, \rho) = -\tau^2 + O(\tau^3, \tau\rho), \quad M(\tau, \rho) = \rho + O(\tau^2\rho, \rho^2).$$

Hence its smooth extension has Whitney-fold normal form and local degree zero. This is distinct from the cross-cap singularity of $S : D \rightarrow \mathbb{R}^3$ in Proposition 6.2.

Proof. From $N = (1 - c)(1 - c - 2u)$ and the explicit rational expression for M in [1], evaluation at $c = 1$, $s = 0$, and $u = 1$ gives $N_t = N_u = M_t = 0$ and $M_u = -1$. Also $N_{tt}(0, 1) = -2$. Taylor expansion gives the displayed normal form. A small source circle maps at leading order to $-\varepsilon^2 \cos^2 \theta + i\varepsilon \sin \theta$, which stays in the closed left half-plane and has winding number zero. $\square$

8.5 Source Topology and Induced-Metric Refinement

Let

$$D_{\text{reg}} := D \setminus \{P_0, P_+, P_-\}, \quad g := S^* \delta_{\mathbb{R}^3}.$$

The latter is a positive-definite metric on D_{reg} .

Proposition 8.9 (The source is not a closed non-orientable surface). *The regular source D_{reg} is a connected orientable surface with boundary. It has*

$$H_1(D_{\text{reg}}; \mathbb{Z}) \cong \mathbb{Z}^2,$$

and its orientation double cover is the disconnected trivial cover $D_{\text{reg}} \sqcup D_{\text{reg}}$. In particular, it is not a closed non-orientable surface.

The equality $S(t, 1) = S(-t, 1)$ does not change this conclusion: the involution $t \mapsto -t$ has the fixed point $t = 0$, so it is not a free orientation-reversing boundary gluing. The sign local system in Proposition 8.6 instead records sheet exchange in the singular image.

Proof. The rectangle D is an oriented disk with boundary. Removing the boundary point P_0 leaves a contractible surface with boundary, and removing the two interior points $P_{\pm}$ gives the standard twice-punctured disk. This has first integral homology $\mathbb{Z}^2$. On D_{reg} the vector field

$$\nu = \frac{S_t \times S_u}{\|S_t \times S_u\|}$$

is a global continuous unit normal, which fixes an orientation and makes the orientation cover trivial. Finally, $S(t, 1) = S(-t, 1)$ follows directly from the evenness of $\cos t$ and $Q(t)$, while $t = 0$ is fixed by the involution. $\square$

Proposition 8.10 (Induced-metric Galerkin consistency). *Let*

$$\mathcal{H}_g = L^2(D_{\text{reg}}, dA_g), \quad \mathcal{E}_g(f, h) = \int_{D_{\text{reg}}} g^{\alpha\beta} \partial_\alpha f \partial_\beta \bar{h} dA_g.$$

For any nested conforming finite-element spaces $V_n \subset V_{n+1} \subset H^1(D_{\text{reg}})$, choose bases satisfying

$$\phi_i^{(n)} = \sum_{\alpha} (J_n)_{\alpha i} \phi_{\alpha}^{(n+1)}.$$

If the induced metric is integrated exactly, the mass and stiffness matrices

$$(G_n)_{ij} = \langle \phi_i^{(n)}, \phi_j^{(n)} \rangle_{\mathcal{H}_g}, \quad (K_n)_{ij} = \mathcal{E}_g(\phi_i^{(n)}, \phi_j^{(n)})$$

satisfy the exact refinement identities

$$J_n^* G_{n+1} J_n = G_n, \quad J_n^* K_{n+1} J_n = K_n.$$

Proof. Both sides are the same L^2 inner product, respectively Dirichlet form, evaluated on the same two functions in V_n after writing them in the basis of V_{n+1} . Equality follows by bilinearity. This is an exact pullback identity, not an asymptotic convergence claim. $\square$

Proposition 8.11 (Zero capacity and fixed-boundary removability). *Let $p \in \{P_0, P_+, P_-\}$ and let U_p be a sufficiently small source neighbourhood, taken as a half-neighbourhood at the boundary point P_0 . For the local 1-capacity*

$$\text{Cap}_{g, U_p}(\{p\}) := \inf_{\eta} \left(\|\eta\|_{L^2(U_p, dA_g)}^2 + \mathcal{E}_g(\eta, \eta) \right),$$

where η ranges over compactly supported smooth cutoffs equal to one near p , one has

$$\text{Cap}_{g, U_p}(\{p\}) = 0.$$

In particular, if $Z_{\text{int}} = \{P_+, P_-\}$ and the outer boundary is held at the Dirichlet condition, then

$$\overline{C_c^\infty(D^\circ \setminus Z_{\text{int}})}^{\mathcal{E}_1} = \overline{C_c^\infty(D^\circ)}^{\mathcal{E}_1}, \quad \mathcal{E}_1(f) := \mathcal{E}_g(f, f) + \|f\|_{L^2(dA_g)}^2. \quad (3)$$

Thus removing the two interior cross-cap points does not alter the fixed-boundary Friedrichs Dirichlet form.

Proof. By Proposition 6.2, each interior germ is right-left equivalent to the standard cross-cap

$$f_{\times}(x, y) = (x, xy, y^2),$$

and P_0 gives its boundary half-germ. Smooth source and target coordinate changes with bounded invertible differentials compare the induced metrics and their energies by fixed positive constants. It therefore suffices to prove the capacity statement for $f_{\times}$; its pullback is a Whitney metric in the sense of [30].

For the standard germ,

$$g_{\times} = \begin{pmatrix} 1 + y^2 & xy \\ xy & x^2 + 4y^2 \end{pmatrix}, \quad dA_{g_{\times}} = w \, dx \, dy, \quad w^2 = x^2 + 4y^2 + 4y^4.$$

Put $\varrho = \|f_{\times}(x, y)\|$. On the regular locus, $|\nabla_{g_{\times}} \varrho| \leq 1$, because $d\varrho(v) = \langle f_{\times}/\varrho, df_{\times}(v) \rangle$. If $0 < r < 1$ and $\varrho < r$, then $|x| < r$ and $|y| < \sqrt{r}$; moreover $w \leq 3\sqrt{r}$. Hence the induced area satisfies

$$A(r) := \text{Area}_{g_{\times}} \{\varrho < r\} \leq 12r^2.$$

For $0 < \varepsilon < 1$, take the logarithmic cutoff

$$\chi_\varepsilon = \begin{cases} 1, & \varrho \leq \varepsilon^2, \\ \frac{\log(\varepsilon/\varrho)}{\log(1/\varepsilon)}, & \varepsilon^2 < \varrho < \varepsilon, \\ 0, & \varrho \geq \varepsilon. \end{cases}$$

After smoothing its two corners without changing the estimates, $\|\chi_\varepsilon\|_{L^2}^2 = O(\varepsilon^2)$. The gradient bound and Stieltjes integration using $A(r) = O(r^2)$ give

$$\mathcal{E}_g(\chi_\varepsilon, \chi_\varepsilon) \leq \frac{1}{\log^2(1/\varepsilon)} \int_{\varepsilon^2 < \varrho < \varepsilon} \frac{dA_g}{\varrho^2} = O\left(\frac{1}{\log(1/\varepsilon)}\right) \rightarrow 0.$$

Restriction to the boundary half-germ can only decrease both terms. This proves zero local capacity at all three points.

Finally, multiply any $\phi \in C_c^\infty(D^\circ)$ by $1 - \chi_\varepsilon$ near each point of Z_{int} . The preceding bounds and the product rule show convergence to ϕ in $\mathcal{E}_1$. The reverse inclusion in (3) is immediate. This is the standard capacity-removability mechanism for Dirichlet forms [31], proved here directly for the cross-cap germ. $\square$

Proposition 8.12 (Uniform front/seam coercivity). *Let $p = P_+$ or P_- , put $\varrho(q) = \|S(q) - S(p)\|$, and let $C_r = \{\varrho = r\}$ in a sufficiently small source neighbourhood. If Π_r denotes the ds_g -average on the connected link C_r , then, with constants independent of r and of the sign of p ,*

$$\int_{0 < \varrho < r_0} \frac{|u - \Pi_\varrho u|^2}{\varrho^2} dA_g \leq C \mathcal{E}_g(u, u), \quad u \in C_c^\infty(\{0 < \varrho < r_0\}). \quad (4)$$

For the standard cross-cap, the two front charts are

$$x = \frac{r \cos \theta}{\sqrt{1 + r \sin \theta}}, \quad y = \epsilon \sqrt{r \sin \theta}, \quad 0 < \theta < \pi, \quad \epsilon \in \{+1, -1\}, \quad (5)$$

and the seam charts are $x = \varsigma \xi$, $y = \xi z$ with $\varsigma \in \{+1, -1\}$. On every fixed seam strip $|z| \leq K$, if

$$W_\xi(z) = \sqrt{1 + 4z^2 + 4\xi^2 z^4}, \quad \bar{u}_\xi = \frac{\int_{-K}^K u(\xi, z) W_\xi(z) dz}{\int_{-K}^K W_\xi(z) dz},$$

then

$$\int_{-K}^K |u - \bar{u}_\xi|^2 \xi^2 W_\xi dz \leq C_K \xi^4 \int_{-K}^K A_\xi^{ij} \partial_i u \partial_j \bar{u} dz, \quad A_\xi = \sqrt{\det g} g^{-1}. \quad (6)$$

Consequently a seam has no independent square-integrable zero channel: it only matches the constant modes of its adjacent front charts.

Proof. For $f_\times(x, y) = (x, xy, y^2)$ one has

$$\varrho^2 = x^2(1 + y^2) + y^4, \quad w^2 = \det g = x^2 + 4y^2 + 4y^4,$$

and direct inversion of g gives

$$1 - |\nabla_g \varrho|^2 = \frac{x^2 y^4}{(x^2 + 4y^2 + 4y^4)(x^2(1 + y^2) + y^4)} \leq \frac{y^2}{4} \leq \frac{\varrho}{4}. \quad (7)$$

The anisotropic path $(x, y) \mapsto (sx, \sqrt{s}y)$ has image velocity $(x, \frac{3}{2}\sqrt{s}xy, y^2)$, so ambient distance and path length give

$$\varrho \leq d_g(0, (x, y)) \leq \frac{3}{2}\varrho. \quad (8)$$

In the front coordinates (5), $\varrho = r$ and, away from the two seam ends,

$$g_{rr} = 1 + O(r), \quad g_{r\theta} = O(r^2), \quad g_{\theta\theta} = r^2(1 + O(r)).$$

Moreover $|\partial_\theta f_\times|/r$ is bounded by $C(1 + \sqrt{r/\sin\theta})$, an integrable majorant. The two front branches therefore join through the seams to one metric circle and satisfy

$$\frac{\text{length}_g(C_r)}{r} \longrightarrow 2\pi, \quad \|\nabla_g \varrho - 1\|_{L^\infty(C_r)} \longrightarrow 0. \quad (9)$$

Wirtinger's inequality on this circle, followed by coarea and (7), proves (4).

In a seam chart the exact identities are

$$\det g = \xi^4 W_\xi^2, \quad A_\xi^{zz} - \frac{|A_\xi^{\xi z}|^2}{A_\xi^{\xi\xi}} = \frac{W_\xi}{\xi^2(1 + 4z^2)}.$$

The Schur-complement bound and the one-dimensional weighted Poincaré inequality on $[-K, K]$ give (6). Its transverse normal operator is

$$-\frac{1}{\sqrt{1 + 4z^2}}\partial_z \left(\frac{1}{\sqrt{1 + 4z^2}}\partial_z \right).$$

The zero solutions are a constant and a primitive of $\sqrt{1 + 4z^2}$; neither is localized in $L^2(\mathbb{R}, \sqrt{1 + 4z^2} dz)$. Thus the seams create no additional zero mode. Finally, Euclidean Whitney coordinates put either actual lateral germ into the same weighted leading form (x, y^2) , with all remaining terms of positive weighted order under $(x, y) \mapsto (rx, \sqrt{r}y)$. Shrinking the neighbourhood absorbs those terms, so the constants can be chosen uniformly for P_+ and P_- . This is the two-chart uniform-estimate mechanism used in weighted conic parametrices; compare [32]. $\square$

Proposition 8.13 (Cross-cap Green parametrix and boundary space). *Fix the outer Dirichlet boundary, let $H_F = -\Delta_g$ be the Friedrichs operator, and let $H_{\min}$ be the graph closure of its restriction to $C_c^\infty(D^\circ \setminus \{P_+, P_-\})$. For $z \in \mathbb{C} \setminus \mathbb{R}$ and $\sigma \in \{+, -\}$ there is a Green vector*

$$G_{\sigma,z} = -\frac{1}{2\pi} \log \varrho_\sigma + R_{\sigma,z}, \quad R_{\sigma,z} \in C^0, \quad (10)$$

where $\varrho_\sigma = \|S - S(P_\sigma)\|$. These vectors give

$$\ker(H_{\min}^* - z) = \text{span}\{G_{+,z}, G_{-,z}\}, \quad n_+(H_{\min}) = n_-(H_{\min}) = 2.$$

Every $u \in \text{Dom}(H_{\min}^*)$ has, modulo $\text{Dom}(H_{\min})$, unique coefficients

$$u = a_\sigma \log \varrho_\sigma + b_\sigma + o(1) \quad \text{at } P_\sigma, \quad (11)$$

and the boundary form is

$$\langle H_{\min}^* u, v \rangle - \langle u, H_{\min}^* v \rangle = 2\pi \sum_{\sigma=\pm} (a_\sigma \bar{d}_\sigma - b_\sigma \bar{c}_\sigma), \quad (12)$$

where (c_σ, d_σ) are the coefficients of v . Hence the self-adjoint extensions form the usual $U(2)$ family of Lagrangian boundary conditions; the Friedrichs member is $a_+ = a_- = 0$.

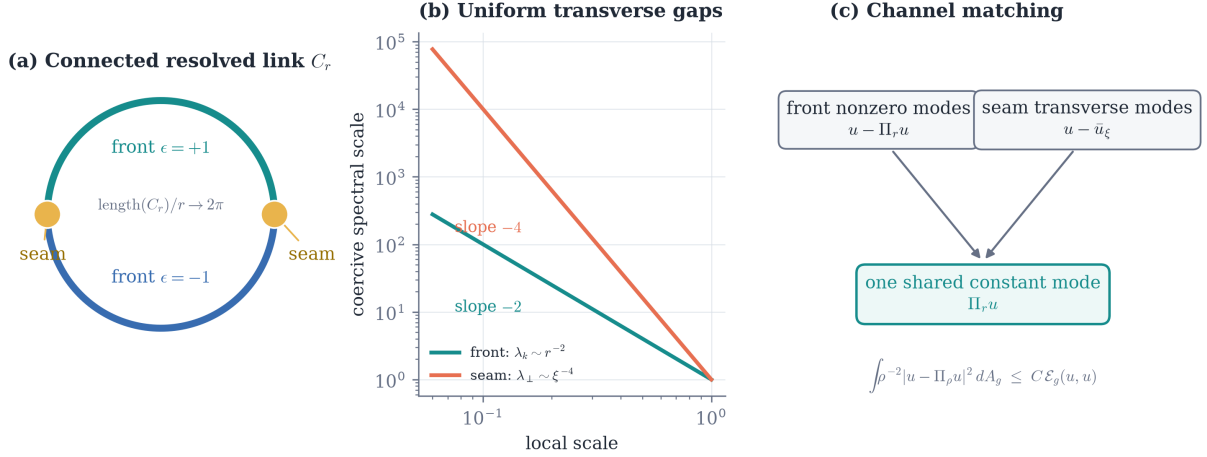


Figure 9: Front–seam decomposition near an interior cross-cap. The two front charts meet through seam charts to give one connected link. Their respective normal scales are r^{-2} and ξ^{-4} ; the uniform estimates leave exactly one matched constant channel and no seam-localized zero mode (Proposition 8.12).

Proof. For the standard germ, direct calculation away from the origin gives

$$\Delta_g \log \varrho = \frac{2x^2y^2(3x^2y^4 + 5x^2y^2 + x^2 + 7y^6 + 5y^4)}{(x^2 + 4y^2 + 4y^4)^2(x^2(1 + y^2) + y^4)^2}. \quad (13)$$

It is $O(r^{-1})$ on closed front subcharts and $O(\xi^{-2})$ in a seam, whose area density is $\xi^2 W_\xi$. Thus it belongs locally to L^p for $1 < p < 3/2$. The positive-weight perturbations of the actual Whitney germs preserve the nonempty range $1 < p < 4/3$. Equations (7) and (9) also give the distributional flux

$$\lim_{r \downarrow 0} \int_{C_r} \partial_\nu \log \varrho \, ds_g = 2\pi.$$

Consequently, for a cutoff χ_σ supported near P_σ ,

$$Q_{\sigma,z} := -\frac{\chi_\sigma}{2\pi} \log \varrho_\sigma, \quad (H_F - z)Q_{\sigma,z} = \delta_{P_\sigma} + q_{\sigma,z}, \quad q_{\sigma,z} \in L^p.$$

To make this a genuine local parametrix, choose a partition of unity between the front and seam charts. On the orthogonal complement of Π_r , invert the front normal family using (4); on a seam, use (6). Their overlap leaves only the same constant link mode. The remaining radial normal operator is

$$-\partial_r^2 - r^{-1}\partial_r,$$

whose two zero solutions are 1 and $\log r$. The chart commutators have positive weighted order and are absorbed by a Neumann series after shrinking the neighbourhood. The resulting estimate, uniform in σ and for z in compact subsets of $\mathbb{C} \setminus \mathbb{R}$, is

$$\|v\|_{C^{0,\alpha}(U_\sigma)} \leq C(\|(H_F - z)v\|_{L^p(U_\sigma)} + \|v\|_{L^p(U_\sigma)}), \quad 0 < \alpha < 2 - \frac{2}{p}.$$

This is precisely where both the front and seam estimates are required; the front chart alone is not uniform at its ends. Applying the Friedrichs resolvent to the remainder gives

$$G_{\sigma,z} = Q_{\sigma,z} - (H_F - z)^{-1}q_{\sigma,z},$$

and proves (10). This local construction is also consistent with general Green-function existence on two-dimensional regular metric-measure spaces [33].

The same estimates exclude every nonconstant angular or seam-localized L^2 defect mode. Subtracting the logarithmic Green vector from a maximal domain element leaves a continuous Friedrichs-domain representative, and its value at the puncture is the coefficient b_σ . Thus the graph closure of functions avoiding P_σ is the kernel of this point trace, one codimension per puncture. For nonreal z , subtraction of the two Green vectors leaves an element of $\ker(H_F - z) = \{0\}$, proving the defect-space and index statements. Finally, Green's identity on $D \setminus \bigcup_\sigma \{\varrho_\sigma < \varepsilon\}$ and (9) give (12); the classification by Lagrangian planes is the standard finite-dimensional boundary-form classification of point interactions [34]. $\square$

Proposition 8.14 (Canonical Markov and no-running-scale selection). *Keep the outer Dirichlet condition fixed. Among the self-adjoint extensions of $H_{\min}$ that generate a submarkovian semigroup, H_F is unique.*

Independently, introduce reference lengths $\ell_\sigma > 0$ in the maximal expansions,

$$u = a_\sigma \log(\varrho_\sigma/\ell_\sigma) + b_\sigma^{(\ell)} + o(1).$$

Under the common change $\ell'_\sigma = e^t \ell_\sigma$, the boundary data obey

$$R_t(a, b) = (a, b + ta), \quad a = (a_+, a_-), \quad b = (b_+, b_-). \quad (14)$$

The only Lagrangian boundary plane invariant under R_t for every $t \in \mathbb{R}$ is

$$\mathcal{L}_F = \{(a, b) : a = 0\},$$

the Friedrichs plane. Consequently every non-Friedrichs extension requires either non-Markovian boundary dynamics or a boundary matrix that runs with the chosen logarithmic reference scale; it is not a parameter-free consequence of the local cross-cap geometry.

Proof. Proposition 8.11 proves that the two punctures have zero relative capacity and that removing them does not change the closed fixed-outer-boundary Dirichlet form. The capacity criterion for Markov uniqueness therefore gives H_F as the unique submarkovian extension [31].

For the second statement, write the Green form on $\mathcal{B} = \mathbb{C}_a^2 \oplus \mathbb{C}_b^2$ as

$$\omega((a, b), (c, d)) = 2\pi(a \cdot \bar{d} - b \cdot \bar{c}).$$

Equation (14) follows directly from $\log(\varrho/\ell) = \log(\varrho/\ell') + t$; this is the two-centre version of the logarithmic reference-length running for planar point interactions [34, Remark 2.11]. If a Lagrangian plane $\mathcal{L}$ is invariant under every R_t , then for each $(a, b) \in \mathcal{L}$ it also contains

$$N(a, b) := \left. \frac{d}{dt} \right|_{t=0} R_t(a, b) = (0, a).$$

Isotropy gives

$$0 = \omega((a, b), N(a, b)) = 2\pi\|a\|^2,$$

so $a = 0$. Hence $\mathcal{L} \subseteq \mathcal{L}_F$; both spaces have complex dimension two, and therefore they are equal. Direct substitution shows that $\mathcal{L}_F$ is invariant. $\square$

Theorem 8.15 (Canonical soft extension from zero-energy Green data). *Let*

$$\mathcal{V}_0 := \overline{C_c^\infty(D^\circ)}^{\mathcal{E}_1} \hookrightarrow \mathcal{H}_g.$$

The embedding is compact, and there is a constant $\lambda_D > 0$ such that

$$\mathcal{E}_g(u, u) \geq \lambda_D \|u\|_{\mathcal{H}_g}^2, \quad u \in \mathcal{V}_0. \quad (15)$$

Consequently $0 \in \rho(H_F)$ and $H_{\min} \geq \lambda_D I$.

The Green construction of Proposition 8.13 therefore extends to $z = 0$. Put

$$h_\sigma := -2\pi G_{\sigma,0}, \quad \sigma \in \{+, -\},$$

and define the zero-energy Robin matrix R_0 by

$$h_\sigma = \delta_{\tau\sigma} \log \varrho_\tau + (R_0)_{\tau\sigma} + o(1) \quad \text{at } P_\tau. \quad (16)$$

Then

$$\ker H_{\min}^* = \text{span}\{h_+, h_-\}, \quad R_0 = R_0^*.$$

The reflection $t \mapsto -t$ gives real numbers r_0, g_0 with

$$R_0 = \begin{pmatrix} r_0 & g_0 \\ g_0 & r_0 \end{pmatrix}. \quad (17)$$

Under the common reference-length change $\ell'_\sigma = e^t \ell_\sigma$, its representative transforms as

$$R_0^{(\ell')} = R_0^{(\ell)} + tI_2; \quad (18)$$

in particular, g_0 is reference-scale invariant.

The Krein–von Neumann extension H_K of $H_{\min}$ is the self-adjoint extension

$$\begin{aligned} \text{dom } H_K &= \text{dom } H_{\min} \dot{+} \ker H_{\min}^* \\ &= \{u \in \text{dom } H_{\min}^* : b = R_0 a\}. \end{aligned} \quad (19)$$

It is nonnegative, has $\ker H_K = \text{span}\{h_+, h_-\}$, and is not submarkovian. Thus the global positive operator selects a canonical non-Friedrichs member without a fitted boundary matrix, although it does not identify source states with a quotient of the singular image.

Proof. Away from the three rank-loss points, compactness is the ordinary Rellich theorem on a finite smooth atlas. Near a rank-loss point, decompose a function into its link average $m(r) = \Pi_r u$ and its angular complement. Equation (4) gives

$$\int_{\varrho < \delta} |u - \Pi_\varrho u|^2 dA_g \leq C\delta^2 \mathcal{E}_g(u, u).$$

Coarea, equations (7) and (9), and the radial part of the energy give

$$\int_0^{r_0} r |m'(r)|^2 dr \leq C \mathcal{E}_g(u, u), \quad \int_0^{r_0} r |m(r)|^2 dr \leq C \|u\|_{\mathcal{H}_g}^2.$$

The weighted one-dimensional Rellich theorem makes the mean channel compact; equation (6) excludes an additional seam channel. A finite partition of unity proves the compact embedding.

If (15) failed, compactness would give an L^2 limit of unit norm for a sequence whose energy tends to zero. Its weak gradient vanishes on the connected regular source, so it is constant, while its fixed outer Dirichlet trace is zero. This contradiction proves (15). It also gives $0 \in \rho(H_F)$ and the same lower bound on the operator restriction $H_{\min}$.

The $z = 0$ parametrix now gives $H_{\min}^* h_\sigma = 0$ and unit logarithmic coefficient at P_σ . If an element of $\ker H_{\min}^*$ has both logarithmic coefficients zero, it belongs to $\text{dom } H_F$ and is killed by H_F , hence vanishes. Thus $h_\pm$ are a basis. Applying the Green form to two kernel elements shows $R_0 = R_0^*$. Reality and the isometric reflection that exchanges P_+ and P_- give (17); the logarithmic identity used in (14) gives (18).

For a densely defined closed strictly positive symmetric operator, the Krein–von Neumann extension is the unique self-adjoint extension whose domain contains the kernel of the adjoint, and its domain is the first line of (19) [35]. The boundary data of $\text{dom } H_{\min}$ vanish, while (16) says that the kernel contribution has $b = R_0 a$; subtraction proves the second line. Since H_F is strictly positive whereas H_K has a two-dimensional kernel, they are distinct. Proposition 8.14 makes H_F the unique submarkovian extension, so H_K is non-Markovian. $\square$

Corollary 8.16 (Intrinsic non-harmonic variational principle). *The positive spectrum of the reduced operator $H_K|_{(\ker H_K)^\perp}$ is the spectrum of the weak buckling problem*

$$\langle H_{\min} v, H_{\min} \phi \rangle = \lambda \langle v, H_{\min} \phi \rangle, \quad \phi \in \text{dom } H_{\min}. \quad (20)$$

Equivalently it is generated by stationary values of

$$\mathcal{A}_\lambda[v] = \frac{1}{2} \|H_{\min} v\|^2 - \frac{\lambda}{2} \langle v, H_{\min} v \rangle. \quad (21)$$

This is a non-harmonic action derived from $H_{\min}$ itself; λ is the spectral Lagrange multiplier, not a fitted interface coupling [36].

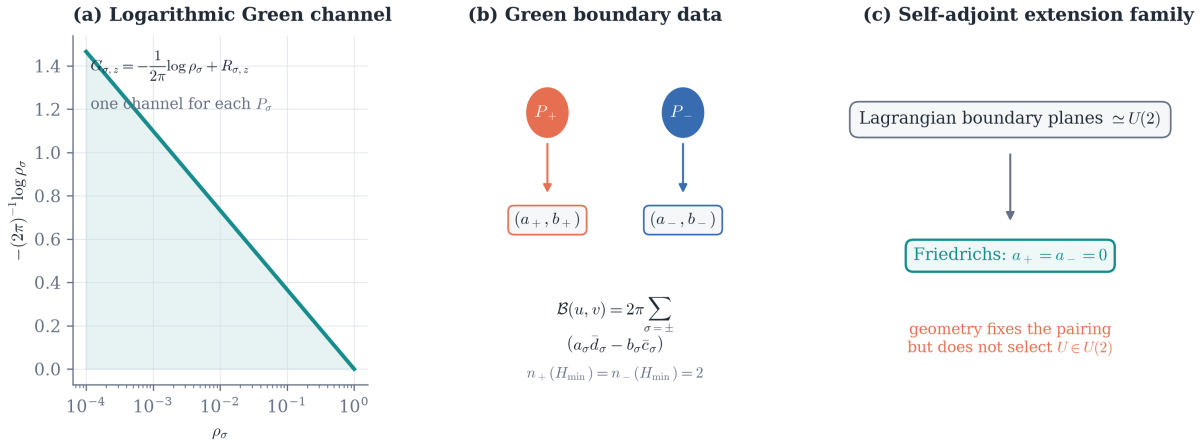


Figure 10: Local Green data and the extension classification at the two interior cross-caps. Each puncture has the logarithmic channel $a_\sigma \log \varrho_\sigma + b_\sigma + o(1)$ and the displayed symplectic Green pairing. Maximal Lagrangian boundary conditions form a $U(2)$ family; the fixed-boundary Friedrichs member sets $a_+ = a_- = 0$ (Proposition 8.13).

Remark 8.17 (What geometry fixes, and what remains external). *The embedded unit circles fix g , dA_g , and the normalized Dirichlet generator through its principal symbol g^{-1} . Thus a kinetic rescaling is fixed once one requires the generator to be exactly Δ_g . The result is compatible with refinement constructions that preserve a discrete de Rham complex [28, 29]; it also supplies the geometry-derived alternative to the earlier counting-Hilbert normalization.*

An arbitrary multiplier of a classical action not tied to this generator remains free. Proposition 8.11 closes the narrower fixed-outer-boundary Markov-form question: the punctures $P_{\pm}$ are removable for the Friedrichs form. Proposition 8.13 shows that the punctured minimal operator is not essentially self-adjoint and classifies its $U(2)$ Green boundary space. The induced geometry fixes the pairing but does not select a non-Friedrichs Lagrangian plane. Proposition 8.14 strengthens this boundary: submarkovianity and parameter-free reference-scale invariance each select the Friedrichs plane uniquely. Theorem 8.15 adds a distinct global statement: strict positivity and the zero-energy Green matrix canonically supply the scale-covariant running matrix of the Krein extension. This does not contradict the local no-go, because R_0 uses the full fixed-boundary resolvent rather than only a cross-cap germ. The point P_0 belongs to the outer boundary and is not an additional interior channel, while the global choice among Dirichlet, Neumann, or Robin outer conditions remains external to the local calculation.

Nor does the induced metric decide whether states live on the source normalization or on a quotient of the singular image. The theorem below shows that the ordinary map from the extension-by-zero normalization defect to the point boundary sheaf is forced to vanish; the nonzero nearby replacement has a different sheet representation. Different extensions can alter propagation through a degenerate locus in related models [37]; that literature is methodological guidance, not a direct theorem about the present cross-caps. In the smooth compact spin setting a Dirac operator can recover the Riemannian metric through its spectral data [38], but the remaining extension data must be fixed before importing such a spectral calibration.

Theorem 8.18 (Normalization-to-Green bridge obstruction). *Let $x_{\sigma} = S(P_{\sigma})$, let $i : Z = \{x_+, x_-\} \hookrightarrow X$, and on pairwise disjoint cross-cap neighbourhoods write*

$$\mathcal{Q}_{\nu, Z, \mathbb{C}} := \bigoplus_{\sigma=\pm} j_{\sigma!} \mathcal{L}_{\text{sgn}, \sigma} \otimes_{\mathbb{Q}} \mathbb{C}. \quad (22)$$

Represent the two logarithmic Green coefficient lines by the skyscraper sheaf

$$\mathcal{G}_{\partial} := i_* \mathbb{C}_Z, \quad \Gamma(Z, \mathbb{C}_Z) = \mathbb{C}_{\text{Green}}^2. \quad (23)$$

Then every ordinary sheaf morphism is zero:

$$\text{Hom}_X(\mathcal{Q}_{\nu, Z, \mathbb{C}}, \mathcal{G}_{\partial}) = 0. \quad (24)$$

The closest nonzero boundary object is instead the nearby/costalk line

$$\mathcal{N}_{\sigma} := H^0(i_{\sigma}^* Rj_{\sigma*} \mathcal{L}_{\text{sgn}, \sigma, \mathbb{C}}) \cong H^1(i_{\sigma}^! j_{\sigma!} \mathcal{L}_{\text{sgn}, \sigma, \mathbb{C}}) \cong \mathbb{C}, \quad (25)$$

so $\mathcal{N}_+ \oplus \mathcal{N}_-$ has the desired dimension two. However, sheet exchange acts as -1 on each $\mathcal{N}_{\sigma}$ and as $+1$ on the scalar logarithmic Green line. Consequently

$$\text{Hom}_{\mathbb{Z}_2}(\mathcal{N}_{\sigma}, \mathbb{C}e_{\sigma}) = 0. \quad (26)$$

If sheet equivariance is discarded and bases are chosen, support-preserving isomorphisms are multiplication by nonzero scalars λ_σ ; reflection relates the two scalars but leaves one nonzero scalar parameter. Requiring an inverse reconstruction map does not remove it, since $\lambda^{-1}\lambda = 1$ for every $\lambda \neq 0$.

Proof. The stalk of $j_{\sigma!}\mathcal{L}_{\text{sgn},\sigma}$ at the pinch is zero by extension by zero, while $\mathcal{G}_{\partial}$ is zero away from the two pinches. Every induced stalk map is therefore zero, and sheaf morphisms are determined by their stalk maps; this proves (24) [20].

Locally the open double stratum of the standard cross-cap is a ray $(0, \varepsilon)$. The recollement triangle for that ray identifies its nearby H^0 line with the degree-one costalk of the extension by zero, equivalently with $H_c^1((0, \varepsilon); \mathcal{L}_{\text{sgn},\sigma,\mathbb{C}})$. This proves (25). By construction the deck exchange of the two normalization sheets acts on this line by -1 .

For $f_{\times}(x, y) = (x, xy, y^2)$, the two preimages of a point on the open double ray are $(0, y)$ and $(0, -y)$. Their ambient radii agree, and hence

$$\log \varrho(0, y) - \log \varrho(0, -y) = 0. \quad (27)$$

Proposition 8.12 shows that this matched constant mode is the only scalar defect channel; there is no independent odd seam mode. Thus the Green coefficient line is the trivial sheet representation. An intertwiner T would obey $T(-v) = T(v)$, which forces $T = 0$ over $\mathbb{C}$ and proves (26). After forgetting this action, maps between the one-dimensional stalks are scalars, and the last statement is immediate. $\square$

Normalization defect and scalar Green-channel obstruction

At a cross-cap pinch, extension by zero has zero stalk; its nearby line and the scalar Green line have opposite sheet parity.

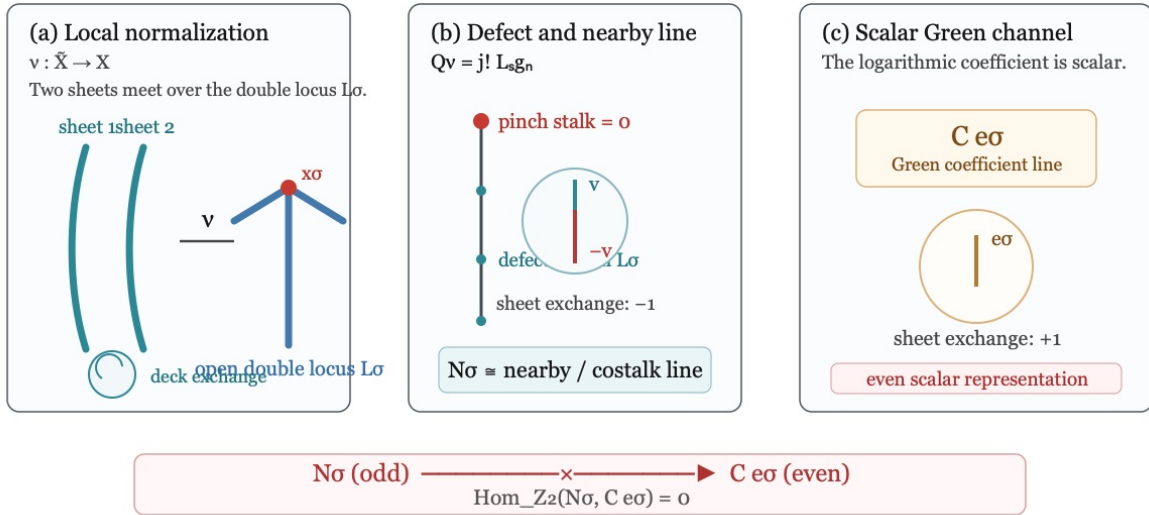


Figure 11: Local normalization-to-Green obstruction at one cross-cap pinch. The extension-by-zero defect has zero pinch stalk, which forces the ordinary sheaf morphism to vanish. Its nonzero nearby/costalk line is sheet-odd, while the scalar logarithmic Green coefficient line is sheet-even; hence the equivariant map also vanishes. This visualizes Theorem 8.18; it does not exclude a bridge based on additional sign-twisted, differential-form, or spinorial structure.

Remark 8.19 (What would reopen the bridge). *The obstruction is specific and local. It does not rule out a bridge to a sign-twisted Green sheaf, a differential-form or spinor operator with a genuine sheet-odd deficiency channel, or a separately oriented/trivialized double cover. Each option adds structure absent from the scalar induced-metric Laplacian. A rescaled odd difference such as $(u(0, y) - u(0, -y))/y$ would also require a chosen transverse coordinate and is not the boundary coefficient classified above. Therefore none may be used as an implicit normalization of the present model.*

8.6 A Derived Phase-Sign Twist and Its Front–Seam Parametrix

Let U_σ be a sufficiently small source neighbourhood of P_σ , $\sigma \in \{+, -\}$, and put $U_\sigma^* = U_\sigma \setminus \{P_\sigma\}$. The phase map $F/|F| : U_\sigma^* \rightarrow S^1$ pulls back the squaring cover $S^1 \rightarrow S^1$, $z \mapsto z^2$. Let

$$\mathcal{E}_{F,\sigma} := \{(q, \zeta) \in U_\sigma^* \times S^1 : \zeta^2 = F(q)/|F(q)|\} \times_{\mathbb{Z}_2} \mathbb{C}_{\text{sgn}} \quad (28)$$

be the associated flat Hermitian sign line.

Proposition 8.20 (Derived sign twist and cone-normal spectrum). *The flat line $\mathcal{E}_{F,\sigma}$ is determined by the existing observation field and has holonomy -1 around either interior cross-cap:*

$$\text{Hol}_{\gamma_\sigma}(\mathcal{E}_{F,\sigma}) = (-1)^{\text{Ind}(P_\sigma)} = -1. \quad (29)$$

No continuous holonomy parameter is introduced.

On the limiting round link from (9), the connection-Laplacian normal family is

$$N(\Delta_{\mathcal{E}_F}) = -\partial_r^2 - r^{-1}\partial_r + r^{-2}D_{1/2}^2, \quad \text{spec}(D_{1/2}) = \mathbb{Z} + \tfrac{1}{2}. \quad (30)$$

Its indicial roots are $\lambda = \pm|n + \frac{1}{2}|$. The two angular weights $n = 0, -1$ have modulus $1/2 < 1$, so both powers $r^{-1/2}$ and $r^{1/2}$ are square-integrable with measure $r dr$. Thus the scalar sign twist has two lowest limit-circle angular channels and does not by itself select the one-dimensional nearby line in (25).

By contrast, restrict the unique spin structure of the oriented source disk to U_σ^ . On either chiral spinor line, its bounding link spin structure has spectrum $\mathbb{Z} + \frac{1}{2}$. Tensoring with $\mathcal{E}_{F,\sigma}$ multiplies the two link holonomies and gives the chiral tangential spectrum*

$$\text{spec}(A_\partial^{F,+}) = \mathbb{Z}, \quad \dim_{\mathbb{C}} \ker A_\partial^{F,+} = 1. \quad (31)$$

For the exact-cone chiral spin-Dirac normal operator

$$N(D_F^+) = c(dr) \left(\partial_r + \frac{1}{2r} + \frac{1}{r} A_\partial^{F,+} \right), \quad (32)$$

the zero angular eigenspinor has radial profile $r^{-1/2}$ and

$$\int_0^\varepsilon |r^{-1/2}|^2 r dr = \varepsilon. \quad (33)$$

On this zero-mode block, the unitary map $f(r) \mapsto r^{-1/2}f(r)\phi_0$ from $L^2(dr)$ to $L^2(r dr)$ conjugates $N(D_F^+)$ to $c(dr)\partial_r$. Its trace $f(0)$ is therefore exactly one complex boundary channel in the chiral exact-cone normal family. The adjoint chiral block has its own one-dimensional trace; consequently the full rank-two spin Dirac operator has a two-dimensional zero angular space, not a single channel. The untwisted bounding spectrum has no eigenvalue in $(-1/2, 1/2)$ and satisfies the usual spinorial Witt gap [22].

Proof. Equation (29) is the monodromy computation of Theorem 4.2: the winding numbers are ± 1 , and either odd winding lifts to the nontrivial deck transformation of the squaring cover. Flat Hermitian lines on a punctured disk are classified by their unitary holonomy, so the fixed value -1 determines this line up to unitary isomorphism.

An anti-periodic section on the unit link has Fourier weights $n + \frac{1}{2}$. Substitution of r^λ in (30) gives $\lambda^2 = (n + \frac{1}{2})^2$; the $L^2(r dr)$ condition is $\lambda > -1$. This proves the scalar statement. The boundary spin holonomy of a spin structure extending over a disk is -1 ; multiplying it by the -1 holonomy of $\mathcal{E}_{F,\sigma}$ gives the periodic spectrum in (31). On its zero eigenspace, (32) reduces to $c(dr)(\partial_r + 1/(2r))$, whose homogeneous solution and norm are exactly (33). The displayed unitary conjugation removes the $1/(2r)$ term. For one chiral block the maximal first-order domain has the endpoint trace $f(0)$ while the minimal domain has zero trace; since its zero eigenspace is one-dimensional, its local quotient has exactly one complex channel. Taking both chiral blocks doubles this dimension. $\square$

Theorem 8.21 (Actual front-seam twisted-spin parametrix). *Let $D_{F,\sigma}^+$ be the chiral spin Dirac operator of the induced metric on U_σ^* , twisted by $\mathcal{E}_{F,\sigma}$, and let $D_{F,\sigma}^-$ be its formal adjoint. If $C_r = \{\varrho_\sigma = r\}$, $L_\sigma(r) = \text{length}_g(C_r)$, and $A_{\sigma,r}^{F,+}$ is the intrinsic twisted chiral Dirac operator on C_r , then*

$$\text{spec}(A_{\sigma,r}^{F,+}) = \frac{2\pi}{L_\sigma(r)}\mathbb{Z}, \quad \dim_{\mathbb{C}} \ker A_{\sigma,r}^{F,+} = 1, \quad \inf_{\lambda \neq 0} |r\lambda| \longrightarrow 1. \quad (34)$$

After the radial density unitary, the front operator has the regular-singular form

$$\mathcal{U}_r D_{F,\sigma}^+ \mathcal{U}_r^{-1} = c(dr) \left(\partial_r + r^{-1} \hat{A}_{\sigma,r}^{F,+} + R_{\sigma,r} \right), \quad \hat{A}_{\sigma,r}^{F,+} := r A_{\sigma,r}^{F,+}, \quad (35)$$

where $rR_{\sigma,r} \rightarrow 0$ in the localized graph norm on closed front subcharts.

In a seam $x = \varsigma\xi$, $y = \xi z$, put

$$\ell_{\sigma,\varsigma}(z) := dF_{P_\sigma}(\varsigma, z), \quad W_0(z) := \sqrt{1 + 4z^2}. \quad (36)$$

The non-degeneracy of dF_{P_σ} implies $\ell_{\sigma,\varsigma}(z) \neq 0$ for every real z . Up to constant unitary changes of frame, the seam normal family is

$$D_{F,\sigma}^+ = \xi^{-2} c_2 B_{\sigma,\varsigma} + O(\xi^{-1}), \quad B_{\sigma,\varsigma} = \frac{1}{W_0(z)} e^{-\frac{i}{2} \arg \ell_{\sigma,\varsigma}} \partial_z e^{\frac{i}{2} \arg \ell_{\sigma,\varsigma}}. \quad (37)$$

Its zero solution has constant modulus and is not in $L^2(\mathbb{R}, W_0(z) dz)$. Hence a seam contributes no localized spinor channel and only transports the unique zero line between its two adjacent front charts.

The front mode inverses, after the density conjugation, are

$$(Q_k f)(r) = \begin{cases} r^{-k} \int_0^r s^k f(s) ds, & k > 0, \\ -r^{-k} \int_r^{r_0} s^k f(s) ds, & k < 0, \\ \int_0^r f(s) ds, & k = 0. \end{cases} \quad (38)$$

Patching these inverses with the seam inverse gives left and right local parametrices with compact positive-weight remainders. After shrinking U_σ , the remainders are removed by a Neumann

series. Consequently

$$\frac{\text{Dom } D_{F,\sigma,\max}^+}{\text{Dom } D_{F,\sigma,\min}^+} \cong \mathbb{C}, \quad \frac{\text{Dom } D_{F,\sigma,\max}}{\text{Dom } D_{F,\sigma,\min}} \cong \mathbb{C}^2, \quad (39)$$

where the first line is the chiral trace represented by $r^{-1/2}\phi_{\sigma,r}$, and the second is the full spin Dirac quotient, with one trace from each chirality.

Proof. For small r , equations (7) and (9) make C_r a smooth circle and give $L_\sigma(r)/r \rightarrow 2\pi$. The source spin structure restricts to the bounding spin structure on C_r , with holonomy -1 , while (29) gives holonomy -1 for $\mathcal{E}_{F,\sigma}$. Their tensor product therefore has trivial holonomy. A unitary connection on a metric circle is determined up to gauge by its holonomy, which proves the exact spectrum and the gap in (34).

For the standard cross-cap, write

$$E = 1 + y^2, \quad w^2 = x^2 + 4y^2 + 4y^4.$$

An exact oriented orthonormal coframe and its Levi-Civita connection are

$$\vartheta^1 = \sqrt{E} dx + \frac{xy}{\sqrt{E}} dy, \quad \vartheta^2 = \frac{w}{\sqrt{E}} dy, \quad \omega_{12} = -\frac{x}{Ew} dy. \quad (40)$$

Substitution of $x = \varsigma\xi$, $y = \xi z$ shows that the transverse unit vector has leading coefficient $\xi^{-2}W_0(z)^{-1}\partial_z$, whereas the spin-connection term is only $O(\xi^{-1})$. Since $F(P_\sigma) = 0$ and dF_{P_σ} is invertible,

$$\frac{1}{2}\partial_z \arg F(\varsigma\xi, \xi z) = \frac{1}{2}\partial_z \arg \ell_{\sigma,\varsigma}(z) + O(\xi).$$

This gives (37). Its kernel is generated by $e^{-i \arg \ell_{\sigma,\varsigma}/2}$, which has modulus one; since $\int_{\mathbb{R}} W_0(z) dz = \infty$, it is not a transverse L^2 mode. The intrinsic Whitney normal form of the actual $P_\pm$ metrics has the same leading weighted coframe, up to constant positive and unitary changes, while the remaining two-jet terms have positive weight [30]. Thus the same conclusion holds at both actual cross-caps.

On a closed front subchart, the metric expansion in the proof of Proposition 8.12, the radial density conjugation, and the flat phase gauge give (35). The identity $(\partial_r + k/r)Q_k f = f$ proves (38). The gap in (34) controls every $k \neq 0$ mode uniformly; the seam calculation shows that patching creates no additional kernel. All chart commutators and metric corrections have positive weighted order, so shrinking the neighbourhood makes their composition with the model inverse have norm less than one. The Neumann correction is the standard first-order regular-singular parametrix step; compare [23]. This comparison is used only after the direct seam calculation, since a Whitney metric is not uniformly conic in the seam coordinates.

Only the $k = 0$ homogeneous constant survives the chiral construction. Undoing the density unitary gives $r^{-1/2}\phi_{\sigma,r}$, which is square-integrable by (33). Its radial coefficient has a maximal H^1 trace and the minimal closure has zero trace, proving the first quotient in (39). The full spinor bundle is the sum of the two chiral lines, so the full quotient is their direct sum and has dimension two. $\square$

Theorem 8.22 (First-order deck-phase comparison and Pin obstruction). *Put $K_\sigma = \ker dS_{P_\sigma}$. Since dF_{P_σ} is invertible,*

$$h_{F,\sigma}(v, w) := \text{Re} \left(dF_{P_\sigma}(v) \overline{dF_{P_\sigma}(w)} \right) \quad (41)$$

is a positive-definite inner product on $T_{P_\sigma}D$ and canonically normalizes the kernel line. For the ruled surface (1), a generator is

$$k_\sigma = \partial_t + \frac{s_\sigma c_0}{2(1-c_0)} \partial_u, \quad s_\sigma := \sin(\sigma t_0), \quad dS_{P_\sigma}(k_\sigma) = 0, \quad (42)$$

and

$$|dF_{P_\sigma}(k_\sigma)|^2 = \frac{655}{4} - 72\sqrt{5} > 0. \quad (43)$$

Let k be either $h_{F,\sigma}$ -unit generator of K_σ , put $a = dF_{P_\sigma}(k) \in S^1$, and choose one α with $\alpha^2 = a$. The two branches of the normalization double locus approach the pinch as $P_\sigma \pm rk + O(r^2)$, so the limiting square-root phases over both branches form

$$\{\alpha, i\alpha, -\alpha, -i\alpha\}. \quad (44)$$

They constitute a principal $\mathbb{Z}_4$ torsor fitting into the nonsplit extension

$$1 \longrightarrow \mathbb{Z}_2^{\text{phase}} \longrightarrow \mathbb{Z}_4 \longrightarrow \mathbb{Z}_2^{\text{sheet}} \longrightarrow 1. \quad (45)$$

Consequently there is no canonical identification of the normalization-sheet $\mathbb{Z}_2$ torsor with the phase square-root torsor that is compatible with $\zeta^2 = F/|F|$.

Moreover, the linearized sheet exchange is a reflection of the oriented source plane. It has no lift in $\text{Spin}(2)$ preserving either chiral line. Each of its two $\text{Pin}(2)$ lifts is Clifford-odd and interchanges the two chiral lines. Thus neither one-dimensional trace in Theorem 8.21 is deck-stable; the full rank-two trace is the minimal spinorial carrier of normalization-sheet exchange. Unit Clifford normalization fixes the norm of this odd exchange operator, but the two central Pin lifts remain distinct.

Proof. At P_σ , direct substitution of $u_0 = (1-c_0)/2$ in S_t and S_u gives (42). Using the derivatives in Corollary 4.4,

$$dF_{P_\sigma}(k_\sigma) = s_\sigma(1-2c_0) - i \frac{s_\sigma(15c_0-4)}{2Q_0(1-c_0)}.$$

The identity $c_0^2 = 3c_0 - 1$ reduces its squared modulus to (43). Invertibility of dF_{P_σ} also proves directly that (41) is positive definite.

In the Whitney normal form, the two preimage branches of the open double ray have tangent directions $\pm k$. Taylor expansion therefore gives $F(P_\sigma \pm rk + O(r^2)) = \pm r dF_{P_\sigma}(k) + O(r^2)$. The roots of the two limiting phases a and $-a$ are exactly the four points in (44). Multiplication by i cycles them. Its square is the nontrivial phase-deck sign, and reduction modulo two records which normalization branch is used. The only order-two element of $\mathbb{Z}_4$ lies in the kernel, so (45) has no group section.

Finally, the cross-cap sheet exchange is $y \mapsto -y$ in source Whitney coordinates and hence reverses orientation. Reflections lift to Pin rather than Spin ; Clifford multiplication by a unit reflecting vector anticommutes with the two-dimensional chirality operator and exchanges its eigenspaces [24]. The two lifts differ by the central sign. This proves the spinorial statements. $\square$

Theorem 8.23 (Green-derived full-spin interface action). *Cut a sufficiently small normalization neighbourhood along the two preimages of its open double locus. For either pairing of corresponding banks, write $L_1, L_2 \simeq L$ for the paired boundary curves, ν_j for their outward unit conormals,*

γ_j for spinor trace, and D_j for the full phase-twisted Dirac operator on the two cut pieces. Their Green form is

$$\begin{aligned}\Omega_L(\Psi, \Phi) &:= \sum_{j=1}^2 (\langle D_j \psi_j, \phi_j \rangle - \langle \psi_j, D_j \phi_j \rangle) \\ &= \int_L \sum_{j=1}^2 \langle c(\nu_j) \gamma_j \psi_j, \gamma_j \phi_j \rangle ds.\end{aligned}\tag{46}$$

Let $U : \mathbb{S}|_{L_1} \rightarrow \mathbb{S}|_{L_2}$ be either unit Pin lift of the bank reflection from Theorem 8.22. It is full-spin, exchanges chirality, and obeys

$$U^* c(\nu_2) U = -c(\nu_1).\tag{47}$$

For a scalar $\lambda \neq 0$, consider the transmission graph

$$\mathcal{L}_{\lambda U} := \{(\psi, \lambda U \psi) : \psi \in H^{1/2}(L_1; \mathbb{S})\}.\tag{48}$$

Then

$$\Omega_L|_{\mathcal{L}_{\lambda U}} = (1 - |\lambda|^2) \int_L \langle c(\nu_1) \cdot, \cdot \rangle ds.\tag{49}$$

Hence the transmission domain is Green-Lagrangian exactly when $|\lambda| = 1$. If λU is additionally required to remain a Pin lift of the fixed sheet reflection, then $\lambda = \pm 1$; there is no continuous interface coupling.

On this domain the canonical real cut action is

$$\begin{aligned}\mathcal{A}_{\text{cut}}[\Psi] &:= \sum_{j=1}^2 \langle \psi_j, D_j \psi_j \rangle + \frac{1}{2} \Omega_L(\gamma \Psi, \gamma \Psi) \\ &= \frac{1}{2} \sum_{j=1}^2 (\langle \psi_j, D_j \psi_j \rangle + \langle D_j \psi_j, \psi_j \rangle).\end{aligned}\tag{50}$$

The coefficient $1/2$ is the unique real coefficient of the Green transgression for which the action is real for arbitrary cut traces. Its boundary variation vanishes precisely on the Lagrangian transmission graph. Thus Green's identity fixes both the transgression coefficient and the modulus of the relative sheet coupling. It does not distinguish the two central Pin signs and does not fix an overall multiplier of the entire action.

Proof. Equation (46) is the Green formula for a Dirac-type operator on a manifold with boundary [25]. For traces $(u, \lambda U u)$ and $(v, \lambda U v)$, equations (46) and (47) give

$$\Omega_L((u, \lambda U u), (v, \lambda U v)) = \int_L \langle (c(\nu_1) + |\lambda|^2 U^* c(\nu_2) U) u, v \rangle ds,$$

which is (49). The graph has half the dimension of the doubled boundary fiber, so isotropy is equivalent to being Lagrangian. Among scalar multiples of a fixed Pin lift, only the central scalars ± 1 remain in the Pin double cover.

Put $x = \sum_j \langle \psi_j, D_j \psi_j \rangle$. Green's identity gives $\Omega_L(\gamma \Psi, \gamma \Psi) = \bar{x} - x$. For a real number a , the imaginary part of $x + a(\bar{x} - x)$ is $(1 - 2a) \text{Im } x$; it vanishes for arbitrary x exactly when $a = 1/2$. This proves (50) and uniqueness of its relative coefficient. Its first variation is

$$\delta \mathcal{A}_{\text{cut}} = 2 \text{Re} \sum_j \langle \delta \psi_j, D_j \psi_j \rangle + \text{Re} \Omega_L(\gamma \delta \Psi, \gamma \Psi).$$

Admissible traces and variations lie in the same Lagrangian graph, so the last term vanishes. The remaining statements follow. $\square$

Remark 8.24 (The remaining operator-level bridge). *Proposition 8.20 derives a genuine operator twist, not an imposed anti-periodic boundary condition. It does not yet complete the normalization-to-Green bridge. The line $\mathcal{E}_{F,\sigma}$ lives on the punctured source and records winding of F , whereas $\mathcal{L}_{\text{sgn},\sigma}$ lives on the image double stratum and records normalization-deck exchange. Theorem 8.22 now shows that their direct identification is not merely absent: the first-order phase lift is a nonsplit $\mathbb{Z}_4$ extension, and sheet exchange forces the two chiral traces to occur together. The observation pullback metric fixes the first-order kernel normalization, while Clifford multiplication fixes the norm of the resulting odd full-spin exchange. Theorem 8.23 now derives the corresponding normalization-cut action and removes every continuous relative interface coefficient. What remains is the central Pin sign and an overall multiplier of the complete action; Green's identity cannot fix either. Thus no absolute mass scale or clock follows at this stage.*

8.7 $SU(2)$ and Arithmetic Scope

Proposition 8.25 (Ambient-versus-Constrained Symmetry). *Let $Y \subset S^3 \simeq SU(2)$ be the nonempty constrained source subset whose projection gives (1). If Y is proper, then it is not invariant under the full left $SU(2)$ action.*

Proof. The left action of $SU(2)$ on itself is transitive. Any nonempty invariant subset therefore contains the orbit of each of its points, hence all of $SU(2)$. $\square$

Remark 8.26 (Noteworthy Representation-Theoretic Observation). *The ambient Hopf fibration and the line bundle $\mathcal{O}(1) \rightarrow \mathbb{CP}^1$ do carry a canonical $SU(2)$ doublet through Borel–Weil theory. It does not follow that the proper constrained source inherits the full action, nor does the doublet fix a dimensional mass scale. The ambient resemblance is retained as a possible representation-theoretic guide, not as a particle assignment.*

Proposition 8.27 (Real-Geometry Deligne/Weil No-Go). *The uncomplexified real signal geometry specifies neither a complex algebraic variety and compactification nor a finite-field model, Frobenius endomorphism, prime power q , or ℓ -adic sheaf. Consequently neither the reflection $\sigma \leftrightarrow 1 - \sigma$ nor the threshold $\sigma = 1/2$ determines a Deligne mixed Hodge structure or a Weil-cohomological weight.*

Proof. Mixed Hodge weights require algebraic or analytic structure beyond an underlying real topological space [11, 12]; arithmetic Weil weights require still different Frobenius data [39]. None is supplied by the value $\sigma = 1/2$. $\square$

Theorem 8.28 (Deligne weights after explicit complexification). *For the explicitly complexified residual fiber $U = E \setminus D$ of Theorem 5.7, Deligne's mixed Hodge structure on $H^1(U; \mathbb{Q})$ has*

$$W_0 = 0, \quad W_1 = H^1(E; \mathbb{Q}), \quad W_2 = H^1(U; \mathbb{Q}),$$

with

$$\dim \text{Gr}_1^W = 2, \quad \dim \text{Gr}_2^W = 1.$$

The weight-one part has Hodge types $(1, 0)$ and $(0, 1)$ and monodromy eigenvalues $-i, i$; the weight-two boundary part has type $(1, 1)$ and eigenvalue 1.

Proof. The Gysin sequence for the two-point boundary gives a short exact sequence of mixed Hodge structures

$$0 \longrightarrow H^1(E; \mathbb{Q}) \longrightarrow H^1(U; \mathbb{Q}) \longrightarrow \tilde{H}^0(D; \mathbb{Q})(-1) \longrightarrow 0$$

[13]. Since E has genus one, the left term is pure of weight one and dimension two. The reduced cohomology of a two-point set is one-dimensional; its Tate twist has weight two and type $(1, 1)$. The monodromy action is the one computed in Theorem 5.7. $\square$

Remark 8.29 (Reconstruction boundary). *The monodromy matrix determines the rank-three local system on a punctured parameter disk. It does not by itself determine the specialization, canonical, and variation maps needed for a full nearby/vanishing-cycle package. It is also distinct from the rank-one normalization defect $\mathcal{Q}_\nu \simeq j_! \mathcal{L}_{\text{sgn}}$.*

Proposition 8.30 (Weil-realization scope). *The single curve $U_{\mathbb{Q}}$ does not define a Weil cohomology theory. Once such a realization theory with localization and Tate twists is specified, it may evaluate the pair (E, D) , and its boundary contribution is the one-dimensional kernel of the sum map from the two points of D .*

Proof. A Weil cohomology is a functorial system on a category of varieties, not data carried by one object [15]. For a fixed theory, localization gives

$$0 \longrightarrow H^1(E) \longrightarrow H^1(U) \longrightarrow H^0(D)(-1) \longrightarrow H^2(E).$$

The last arrow sums the two point classes, so its kernel is one-dimensional. No new cohomology theory is inferred. $\square$

9 Signal and Physical Observation Points

The similarities in this section are retained because they may guide a future model. Each subsection first states what the present mathematics actually forces and then separates the remaining observation from a physical claim.

9.1 Geometric Ratio and Its Scale Obstruction

We first audit the units before assigning any physical interpretation.

Proposition 9.1 (Scale of the Proposed Geometric Ratio). *Let the ambient three-sphere have radius R and let $r = \varphi^{-2}R$. Then*

$$\eta := \frac{2\pi^2 R^3}{\pi r^2} = 2\pi\varphi^4 R.$$

Thus η has units of length and cannot be a dimensionless mass hierarchy. The normalized quantity

$$\hat{\eta} := \frac{\eta}{R} = 2\pi\varphi^4$$

is dimensionless, but the present geometry supplies no rule turning it into a particle mass or a mass ratio.

Proof. Substitution of $r^2 = \varphi^{-4}R^2$ gives the formula and its scaling under $R \mapsto \lambda R$. Setting $R = 1$ selects units; it does not remove the dimensional origin of η . $\square$

Remark 9.2 (Noteworthy Mass-Hierarchy Observation). *The pure number $2\pi\varphi^4$ and its powers can generate visible numerical hierarchies. This is worth retaining as a geometric pattern. A physical prediction would additionally require an independently derived mass operator, a state-selection rule, and a normalization scale; none is inferred from the numerical resemblance.*

9.2 Observation Coordinates and Poisson Commutation

Proposition 9.3 (No Induced Non-Commutativity for the Power Coordinates). *For the standard Kähler form on $\mathbb{C}^2$, the two power coordinates $X = |z_1|^2$ and $Y = |z_2|^2$ satisfy*

$$\{X, Y\} = 0.$$

Consequently the momentum projection used in this paper does not by itself induce a non-commutative algebra generated by X and Y .

Proof. X depends only on $(z_1, \bar{z}_1)$ and Y only on $(z_2, \bar{z}_2)$. Every mixed derivative in the canonical Poisson bracket therefore vanishes. $\square$

Remark 9.4 (Noteworthy Non-Commutative Direction). *The three components of the full $SU(2)$ moment map instead obey the $\mathfrak{su}(2)$ Poisson relations, so a deformation-quantization programme could begin there. It would be a different observable algebra and would still need an independently fixed action and scale. The resemblance to synchronized signal phases is therefore a research direction, not a result of the present (X, Y) projection.*

9.3 Sign Holonomy and Spectral Asymmetry

Proposition 9.5 (Vanishing Eta Invariant for Sign Holonomy). *Let $D_a = -i d/d\theta$ on the circle with boundary condition $\psi(\theta + 2\pi) = e^{2\pi i a} \psi(\theta)$. Its spectrum is $\{n + a : n \in \mathbb{Z}\}$. For the sign holonomy $a = 1/2$ associated with a single-loop sign reversal,*

$$\text{spec}(D_{1/2}) = \{n + \tfrac{1}{2} : n \in \mathbb{Z}\}, \quad \eta(D_{1/2}) = 0.$$

Proof. The eigenfunctions are $e^{i(n+a)\theta}$. At $a = 1/2$, the involution $n \mapsto -n - 1$ pairs every eigenvalue with its negative, so the defining eta series cancels for $\Re s$ large and its meromorphic continuation vanishes at $s = 0$ [21]. $\square$

Remark 9.6 (Noteworthy Spinorial Observation). *The 4π restoration of Theorem 4.2 and the sign boundary condition above have the same elementary double-cover pattern as spinorial holonomy. The canonical one-dimensional spectral model is nevertheless symmetric and produces no net chirality. A non-zero eta invariant would require additional asymmetric boundary or operator data and is not claimed.*

10 Conclusion

The revised core now separates four levels of statement. First, the geometric results are intrinsic to the explicit maps: the observation field has a topological dipole, its square root has 4π restoration, the surface rank-loss germs satisfy the cross-cap determinant criterion, and the polar observation singularity is a fold. Second, the local fold and cross-cap residuals have the exact sublevel exponent $3/4$, giving a logarithmic resolution cost without invoking a channel model. After an explicit polynomial complexification, the cross-cap residual is an A_3 germ whose Milnor fiber is a twice-punctured genus-one curve. Its finite monodromy and two-step Deligne weight filtration are then computable; neither is inferred from the real geometry alone. Independently on the real slice, every positive-radius cross-cap link is a circle whose image is a transverse figure-eight quotient, while its normalization lift is embedded. On the chosen A_3 completion, the uniquely primitive weights $(2, 1)$ produce an embedded phase orbit whose rectangular shadow is a Lissajous figure-eight and whose missing quadrature coordinate reconstructs the source phase exactly. For the actual ruled-surface germs, however, the canonical Whitney two-jet coefficients satisfy $b^2 + ac^2 = 0$, so their principal-curvature exceptional traces are cusps. This parameter-free calculation closes the proposed direct principal-curvature bridge to either figure-eight construction.

Third, the analytic completion is independent of the geometry: the half-density is fixed by the Haar-to-Lebesgue unitary, the Dirichlet–Hardy space singles out $\Re s = 1/2$ as its point-evaluation boundary, and the phasor Gram limit is precisely Bohr–Haar character orthogonality. The cross-cap normalization also yields a canonical constructible defect sheaf, replacing the earlier unsupported structure-sheaf and exceptional-pullback claims.

At the source level, the parameter domain is an orientable surface with boundary, not a closed non-orientable surface. Its nontrivial local sign behaviour belongs to the normalization defect of the singular image. On its regular locus, the pullback Euclidean metric fixes a scalar L^2 form and a normalized Dirichlet form. Nested conforming Galerkin spaces then have exact mass- and stiffness-matrix pullback identities. This gives a reproducible refinement-compatible operator construction without fitting a Hilbert metric. The rank-loss points have zero local Dirichlet capacity; with the outer Dirichlet boundary fixed, puncturing $P_{\pm}$ leaves the Friedrichs form unchanged. Uniform front/seam estimates nevertheless show that the punctured minimal Laplacian is not essentially self-adjoint: there is exactly one logarithmic channel per interior cross-cap, the deficiency indices are $(2, 2)$, and the Green form gives a $U(2)$ extension family. Neither this classification nor the induced metric alone chooses a non-Friedrichs member. However, zero capacity makes the Friedrichs extension the unique submarkovian member, and it is also the only Lagrangian boundary plane invariant under logarithmic reference-scale changes without a running coupling. Thus every non-Friedrichs point coupling needs scale-covariant boundary data. Compactness and the fixed outer Dirichlet trace make the minimal operator strictly positive; its zero-energy Green matrix supplies such data canonically for the Krein–von Neumann extension. The resulting soft extension is non-Markovian, has two zero modes, and its reduced positive spectrum follows from the parameter-free buckling action. This operator-level selection does not yet identify the point boundary space with the normalization defect of the singular image. The direct scalar identification is now excluded: extension by zero has no pinch stalk, while its nearby/costalk line is sheet-odd and the unique scalar Green defect channel is sheet-even. The phase monodromy of $\sqrt{F}$ nevertheless derives a flat sign line without a holonomy parameter. Its exact-cone scalar normal family is still two-channel, while its twist of the disk-bounding spin Dirac normal family has one zero angular channel per chirality. The actual front/seam parametrix

now gives one chiral trace line and two trace coefficients for the full Dirac operator. The first-order deck–phase comparison fixes the kernel-line norm but produces a nonsplit $\mathbb{Z}_4$ lift; the sheet reflection is Pin-odd and exchanges chirality. Thus a direct one-chiral torsor identification is now excluded. The cut Dirac Green form instead derives a full-spin Lagrangian transmission action, fixes its transgression coefficient to $1/2$, and removes every continuous relative sheet coupling. Only the central Pin sign, overall action multiplier, and global clock remain unfixed.

Finally, the negative results delimit physical interpretation. The proposed volume-to-area quantity scales as a length; the power coordinates Poisson commute; the natural sign-holonomy circle operator has eta invariant zero; and a proper constrained source cannot inherit the full transitive left $SU(2)$ action. The weighted phase geometry fixes the ratio $2 : 1$, but its Euclidean quadratic action minimizes at a degenerate one-channel orbit and does not fix the amplitudes or an absolute clock. These are not failures of the geometric model: they identify the additional operator, dynamics, or representation data that any physical extension must independently derive.

The repetitions of $\sqrt{5}$, the 4π double-cover pattern, the half-density line, ambient $SU(2)$ representation theory, and arithmetic phase decorrelation remain noteworthy observation points. The figure-eight shadow is likewise an exact phase projection, not yet a spacetime wave equation or a physical scattering graph. None is promoted here to a capacity law, particle mass, net chirality, a Deligne weight derived from $\sigma = 1/2$, or a Weil-cohomological identification.

Computational verification. The exact SymPy check `verify_gir_figure_eight.py` covers the new link, lift, phase orbit, and action identities, together with the actual $P_{\pm}$ two-jet coefficients and cusp discriminant. Prime-exponent character orthogonality, the beta-integral asymptotic, and the constructible normalization defect are already established above. Theorem 8.15 and Corollary 8.16 supply a canonical non-Markovian point extension and its non-harmonic variational principle. Theorem 8.18 now closes the proposed scalar normalization-to-Green bridge: the ordinary sheaf map is zero and the nearby/costalk replacement has the wrong sheet representation. Proposition 8.20 derives a phase-sign flat line and identifies a one-channel twisted-spin exact-cone normal family for each chiral block. Theorem 8.21 proves the actual front/seam parametrix and distinguishes the one-dimensional chiral trace from the two-dimensional full-Dirac trace. Theorem 8.22 verifies the actual kernel direction, fixes its observation-pullback norm, and proves that the phase/deck comparison is a nonsplit $\mathbb{Z}_4$ extension whose Pin lift exchanges chirality. Theorem 8.23 derives the full-spin cut action from the Dirac Green form, fixes the real boundary transgression coefficient to $1/2$, and fixes the transmission multiplier to unit modulus, or to ± 1 inside the Pin double cover. The exact script `verify_crosscap_green_normal.py` checks the front metric and flux, source-coordinate radial identities, seam Schur complement, logarithmic residual, formal boundary form, direct $P_{\pm}$ link convergence, sign-twisted indicial roots, and the spin-link spectral shift. The exact script `verify_ruled_topology.py` checks the source topology, the standard cross-cap metric determinant, and the positive-definite quadratic zeros of the actual metric determinant at $P_0, P_{\pm}$; `verify_gir_refinement_connection.py` separately checks the first mass/stiffness pullback identity.

Table 1: Claim-status ledger after the independent structural audit.

Claim	Reference	Status
Topological dipole $\text{Ind}(P_{\pm}) = \mp 1$	Thm. 4.1	Proven geometry
4π restoration of $\sqrt{F}$	Thm. 4.2	Proven monodromy
Surface rank-loss germs are cross-caps	Prop. 6.2	Proven geometry
Polar zero of Ψ is a fold of degree zero	Prop. 8.8	Proven geometry
Regular source is orientable with boundary, not closed non-orientable	Prop. 8.9	Proven source topology
Induced-metric mass/stiffness pullback under conforming refinement	Prop. 8.10	Proven given exact integration
Each rank-loss point has zero local Dirichlet capacity	Prop. 8.11	Proven local analysis
Fixed-outer-boundary Friedrichs form is unchanged by removing $P_{\pm}$	Prop. 8.11	Proven form removability
Uniform front/seam coercivity and one matched constant channel	Prop. 8.12	Proven local analysis
Non-Markovian self-adjoint extensions and Green pairing at $P_{\pm}$	Prop. 8.13	Proven operator classification
Unique submarkovian and no-running-scale extension is Friedrichs	Prop. 8.14	Proven selection/no-go
Strict positivity and canonical Krein-von Neumann extension	Thm. 8.15	Proven global selection
Reduced Krein spectrum obeys the intrinsic buckling action	Cor. 8.16	Proven variational principle
Scalar normalization-to-Green bridge	Thm. 8.18	Proven support/parity no-go
Phase-derived sign line and twisted-spin cone normal channel	Prop. 8.20	Proven chiral normal family
Actual twisted-spin front/seam parametrix and domain trace	Thm. 8.21	Proven; chiral rank 1, full rank 2 per pinch
First-order phase/deck torsor comparison and chirality selection	Thm. 8.22	Proven $\mathbb{Z}_4$ /Pin obstruction; full spin required
Green-derived full-spin normalization-cut action	Thm. 8.23	Proven; transgression 1/2, relative coupling $ \lambda = 1$
$ \mathcal{P}'(c_0) = \sqrt{5}$ for the chosen monic polynomial	Lem. 5.1	Proven; normalization-dependent
Fold/cross-cap sublevel exponent $3/4$	Thm. 5.3	Proven local asymptotic
Residual geometry does not fix channel capacity	Prop. 5.5	Proven no-go
$K_{\times}$ is an A_3 germ with $\mu = \tau = 3$	Thm. 5.6	Proven after complexification
Milnor fiber and spectrum $\{-i, 1, i\}$	Thm. 5.7	Proven after complexification

Claim	Reference	Status
Positive-radius cross-cap link is a transverse figure-eight	Prop. 5.8	Proven real link geometry
A_3 weighted orbit has a 2 : 1 figure-eight shadow	Thm. 5.9	Proven phase geometry
Euclidean phase action does not fix amplitudes or clock	Prop. 5.10	Proven no-go
Actual $P_{\pm}$ principal-curvature traces are cusps	Prop. 5.12	Proven two-jet no-go
Deligne graded dimensions (2, 1)	Thm. 8.28	Proven for $H^1(U)$
Formal-adjoint threshold and full-line half-density	Thm. 7.1, Prop. 8.1	Proven; half-line not self-adjoint
Finite-time Gram bound and Bohr–Haar limit	Thm. 7.4, 8.4	Proven analysis
Dirichlet–Hardy evaluation boundary $\Re s = 1/2$	Thm. 8.2	Proven analysis
Cross-cap normalization defect $\mathcal{Q}_{\nu}$	Prop. 8.6	Proven local sheaf result
$\eta = 2\pi\varphi^4 R$ is not dimensionless	Prop. 9.1	Proven obstruction
Power coordinates satisfy $\{X, Y\} = 0$	Prop. 9.3	Proven obstruction
Sign-holonomy circle model has $\eta(D_{1/2}) = 0$	Prop. 9.5	Proven obstruction
Proper constrained source is not full-left- $SU(2)$ invariant	Prop. 8.25	Proven scope limit
Deligne/Weil identification from uncomplexified real data	Prop. 8.27	Not derivable
Existing Weil realization of (E, D)	Prop. 8.30	Conditional on chosen theory
Capacity, mass, chirality, and particle interpretations	Remarks in §9	Noteworthy observations only

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